Introduction to Machine Learning in Healthcare

Model Evaluation and Optimization

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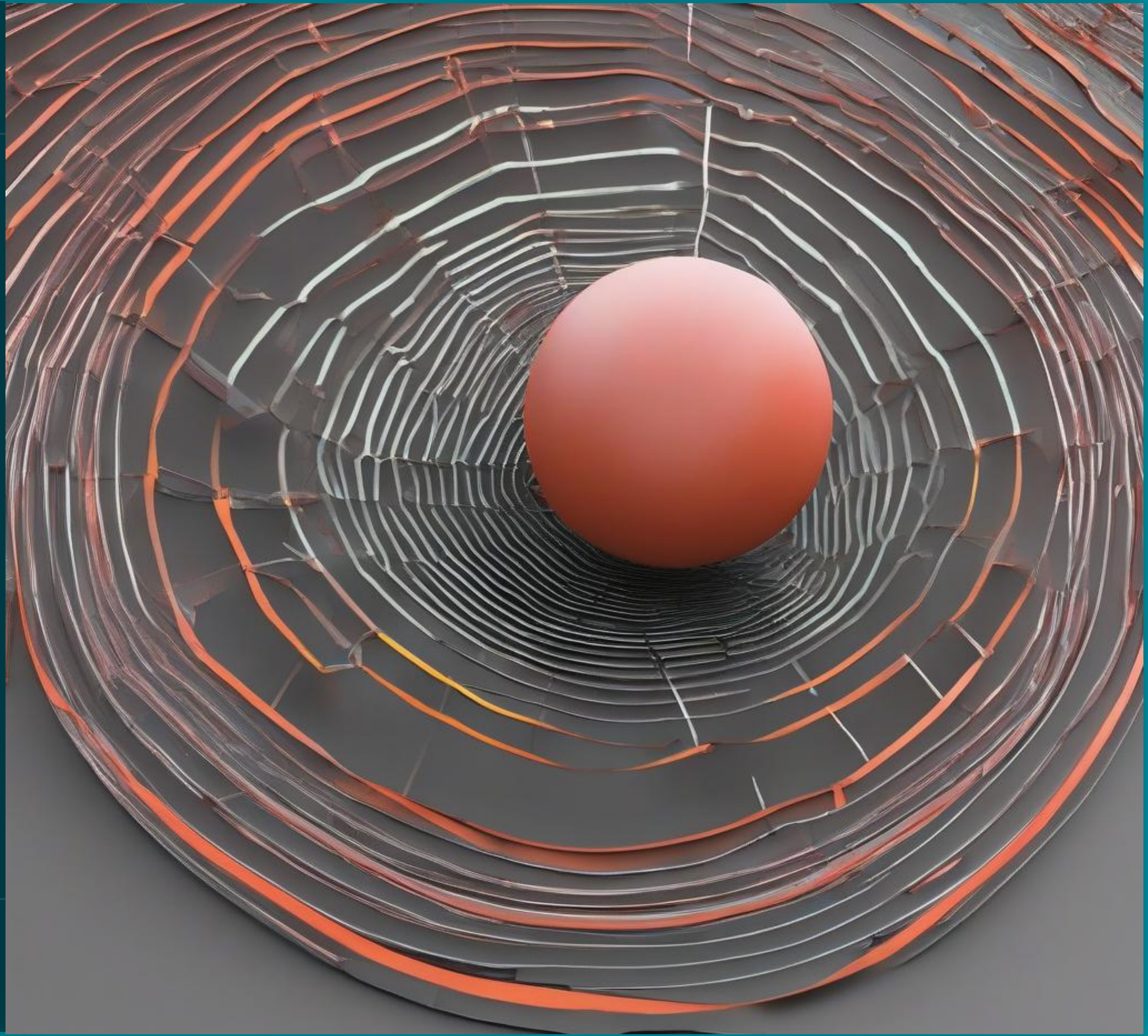

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Content

01

Model Evaluation Metrics for Classification

02

Model Evaluation Metrics for Regression

03

Train-Test Split and Cross-Validation

04

Model Tuning and Optimization

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RELEVANCE TO MEDICINE

Resource Allocation

Patient Safety

Accuracy and Reliability of Diagnoses

Personalized Medicine

KEY TERMS AND CONCEPTS

- Classification
- Regression
- accuracy, precision, recall, F1 score, RMSE, MAE, etc.
- Train-Test Split
- Hyperparameters

01

MODEL EVALUATION METRICS FOR CLASSIFICATION

INTRODUCTION TO MODEL EVALUATION METRICS

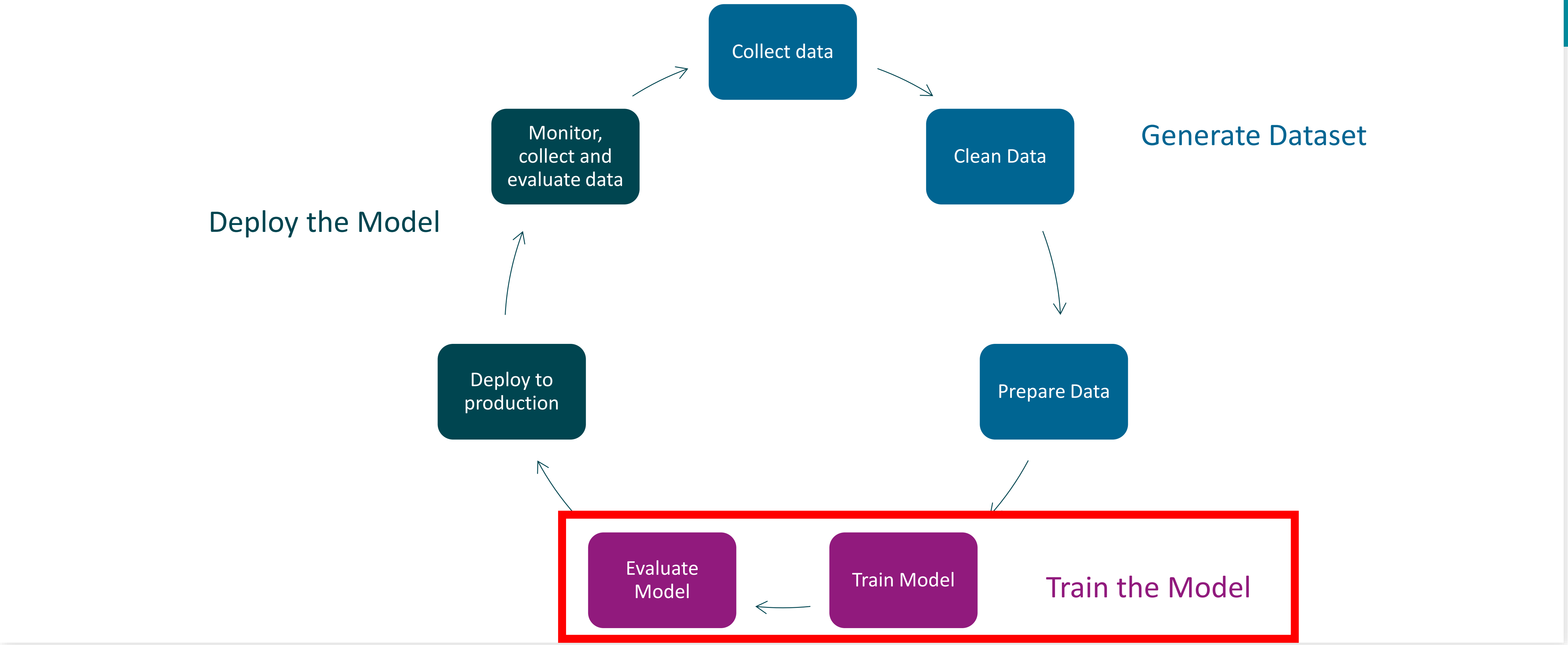
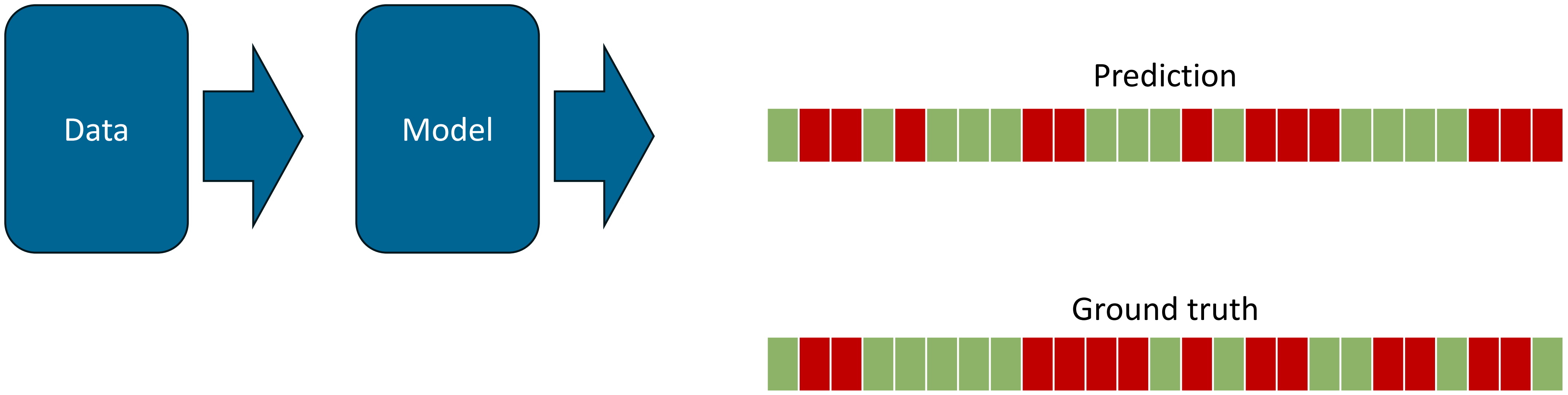


Image from: What is a Machine Learning Pipeline? - Datatron

MODEL OUTPUT

Classification model



CONFUSION MATRIX

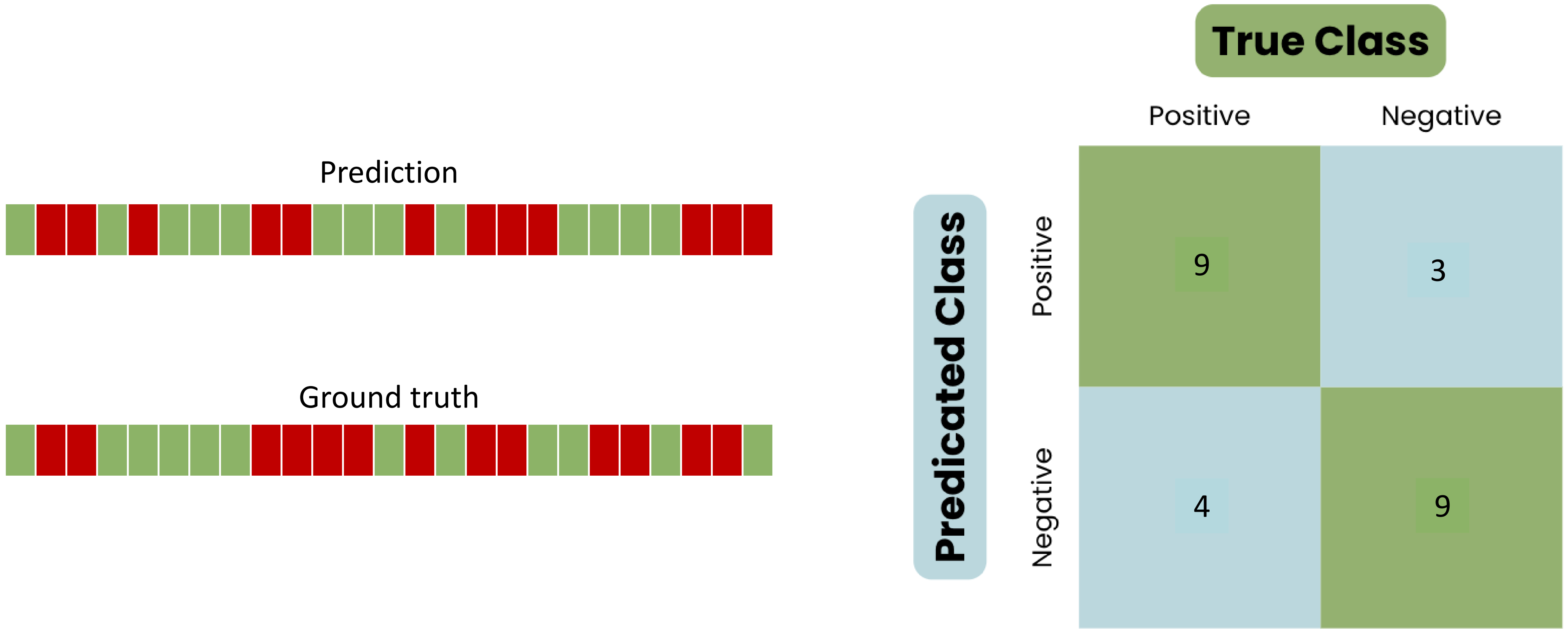


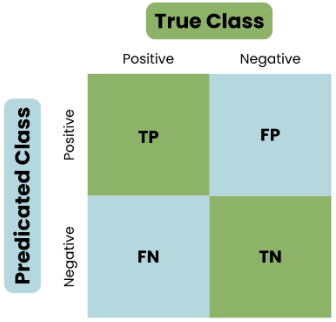
Image from: What is A Confusion Matrix in Machine Learning? The Model Evaluation Tool Explained | DataCamp

ACCURACY

measures the **proportion of true results** (both true positives and true negatives) among the total number of cases examined

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$



ACCURACY

Main limitation

Class imbalance

		True Class	
		Positive	Negative
Predicated Class	Positive	95	5
	Negative	0	0

$$Accuracy = \frac{95 + 0}{95 + 0 + 5 + 0} = 0.95$$

CONFUSION MATRIX

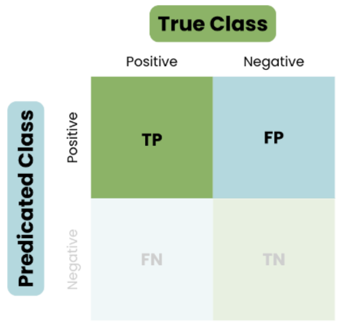
		Predicted Class		
		Positive	**Negative**	
Actual Class	**Positive**	True Positive (TP)	False Negative (FN) **Type II Error**	**Sensitivity** $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) **Type I Error**	True Negative (TN)	**Specificity** $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	**Negative Predictive Value** $\frac{TN}{(TN + FN)}$	**Accuracy** $\frac{TP + TN}{(TP + TN + FP + FN)}$

Image from: What is a Confusion Matrix? | Machine Learning Glossary | Encord | Encord

PRECISION

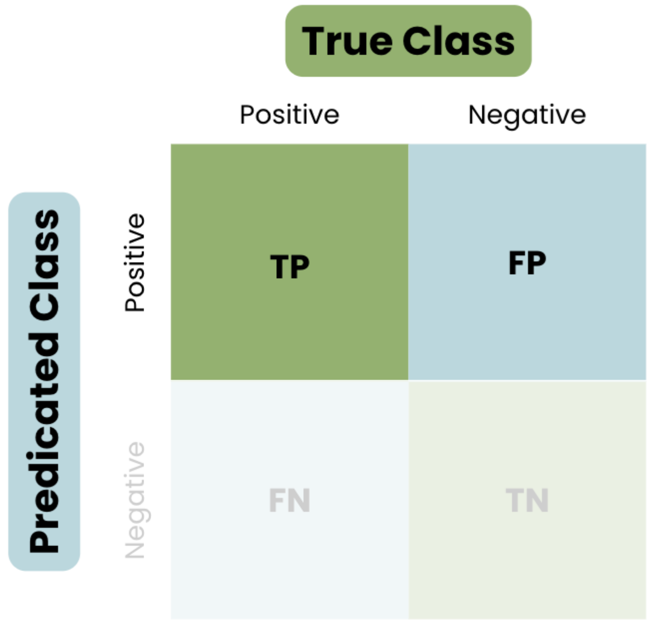
measures the accuracy of the positive predictions

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$



PRECISION

Main limitation



Precision does not consider the false negatives, so it does not provide information about the ability of the model to identify all actual positives

PRECISION

Case study

Case Study Example

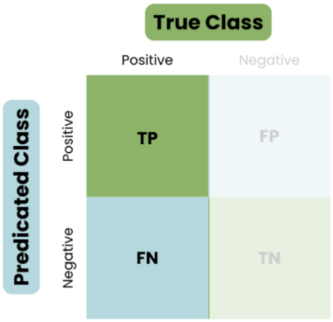
Consider a model designed to predict whether patients have a particular rare genetic disorder. High precision is crucial because:

- False positives could lead to expensive and invasive follow-up tests.
- Ensuring those diagnosed are highly likely to have the disorder can help in focusing medical resources effectively.

RECALL (SENSITIVITY)

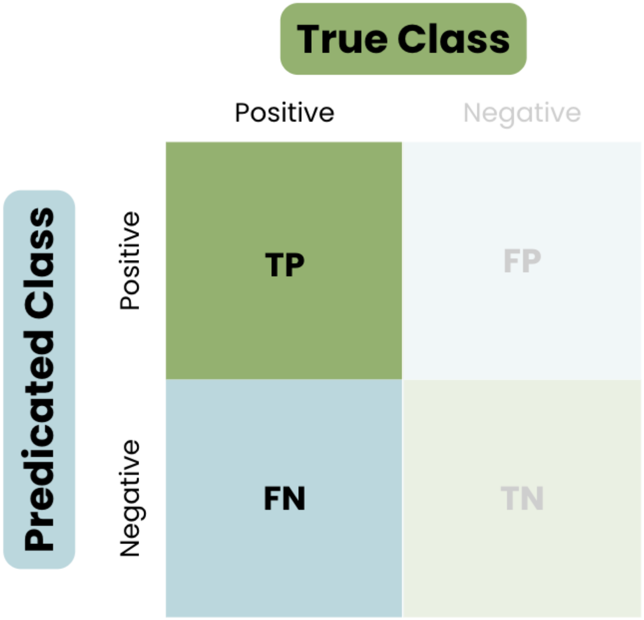
quantifies the ability of the model to find all relevant cases within a dataset

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$



RECALL (SENSITIVITY)

Main limitation



Recall does not consider the false positives, so it doesn't provide information about the accuracy of the positive predictions

RECALL (SENSITIVITY)

Case study

Case Study Example

Consider a model designed to predict whether patients have a highly infectious disease. High recall is crucial because:

- Missing actual cases (false negatives) could result in the spread of the disease.
- Early identification of all infected individuals allows for prompt isolation and treatment to prevent further transmission.

F1 SCORE

is the harmonic mean of
precision and recall

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1 SCORE

Main limitation

Interpretability

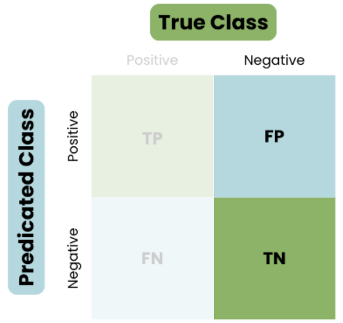
Class Imbalance Sensitivity

SPECIFICITY

True Negative Rate (TNR)

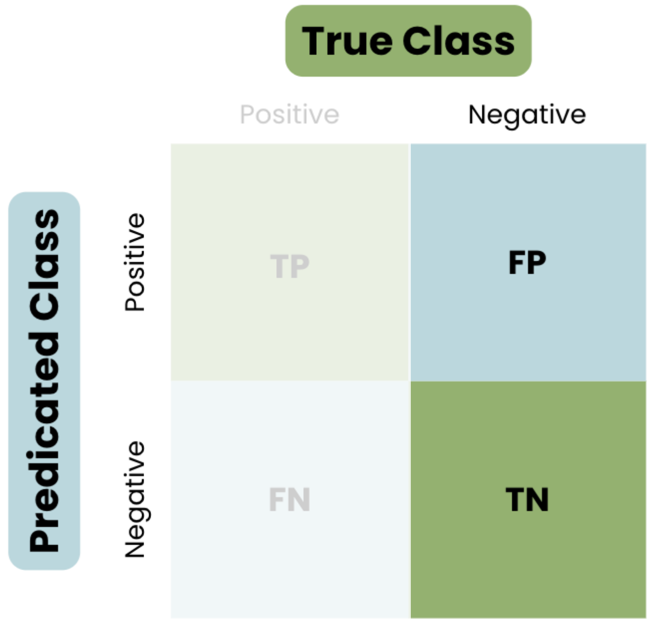
quantifies the ability of the model to correctly identify negative cases

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}}$$



SPECIFICITY

Main limitation



Specificity does not consider the false negatives, so it doesn't provide information about the model's ability to identify all actual positive instances

SPECIFICITY

True Negative Rate (TNR)

Case Study Example

Consider a model designed to screen for a non-life-threatening condition that has a simple but expensive treatment. High specificity is crucial because:

- False positives could lead to unnecessary and costly treatments.
- Ensuring that patients who do not need the treatment are not incorrectly identified as needing it helps in managing healthcare costs and resources effectively.

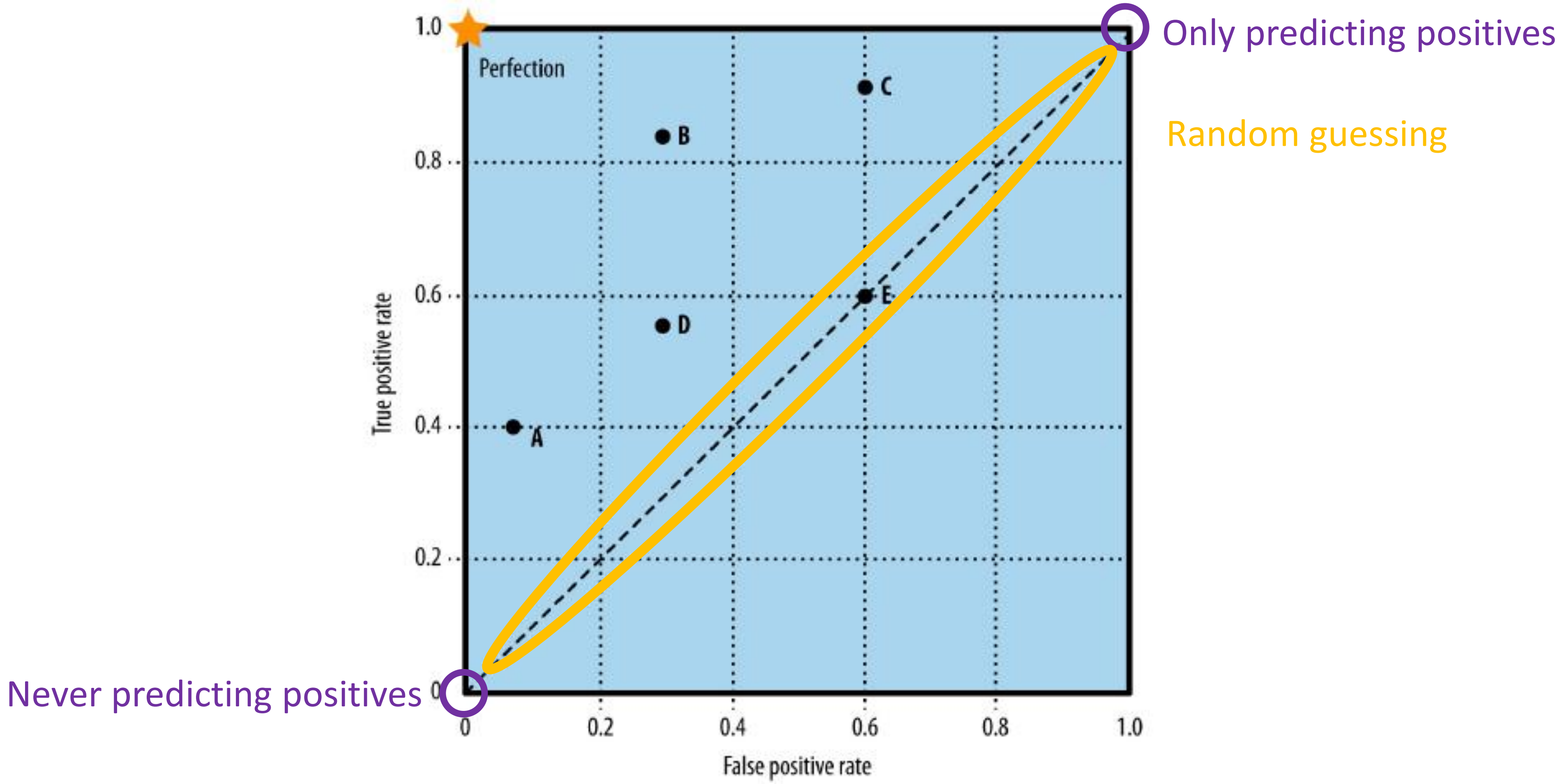
ROC CURVE

used to evaluate the performance of a **binary classification** model

trade-off between **sensitivity** (true positive rate) and **1-specificity** (false positive rate) at **various threshold** settings

helps in **visualizing** the ability of the classifier to distinguish between positive and negative instances

ROC CURVE



Only predicting positives
Random guessing
Never predicting positives
Perfection
1.0
0.8
0.6
0.4
0.2
0
True positive rate
0
0.2
0.4
0.6
0.8
1.0
False positive rate
A
B
C
D
E

ROC CURVE

Instance	True Label	Prediction Score
1	P	0,99
2	P	0,98
3	N	0,96
4	P	0,90
5	N	0,88
6	P	0,87
7	P	0,85
8	P	0,84
9	N	0,80
10	P	0,75
11	N	0,74
12	P	0,71
13	P	0,65

0	0
100	100

1	0
99	100

2	0
98	100

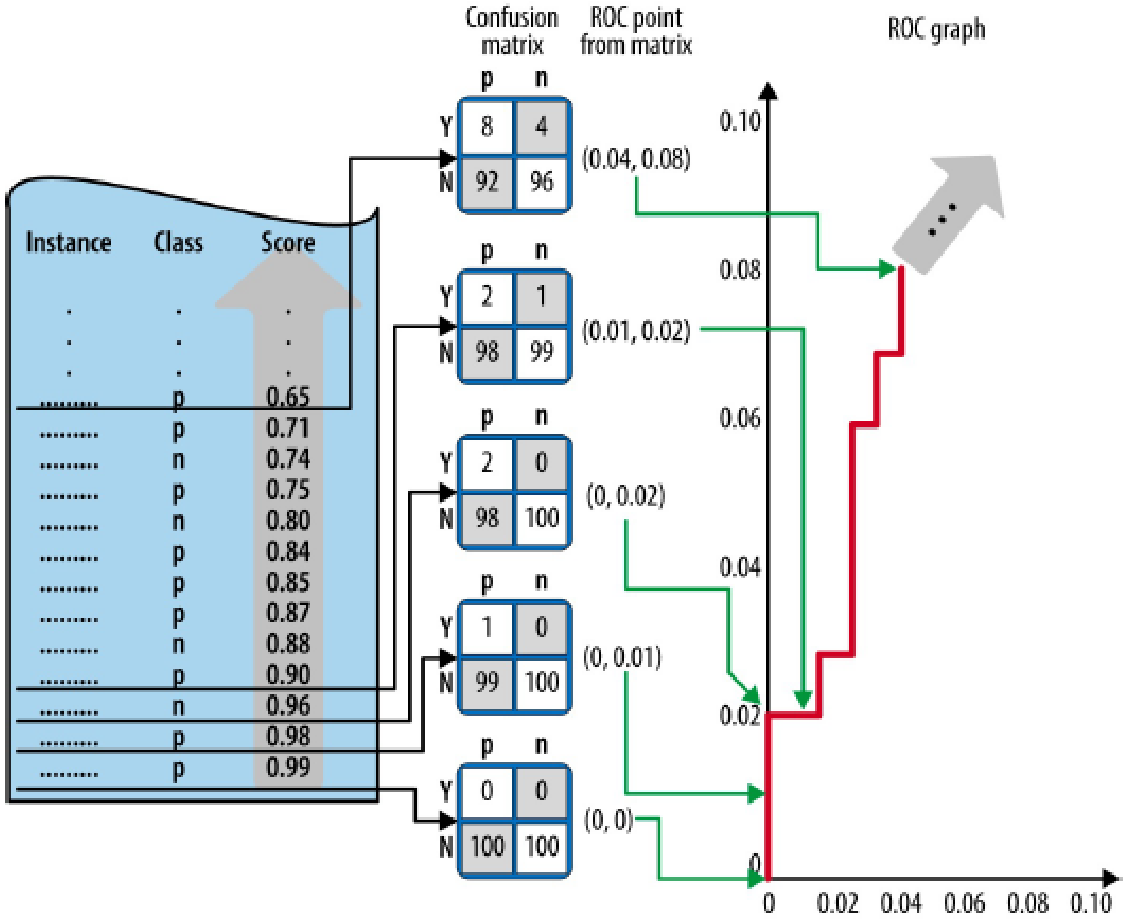
2	1
98	99

...

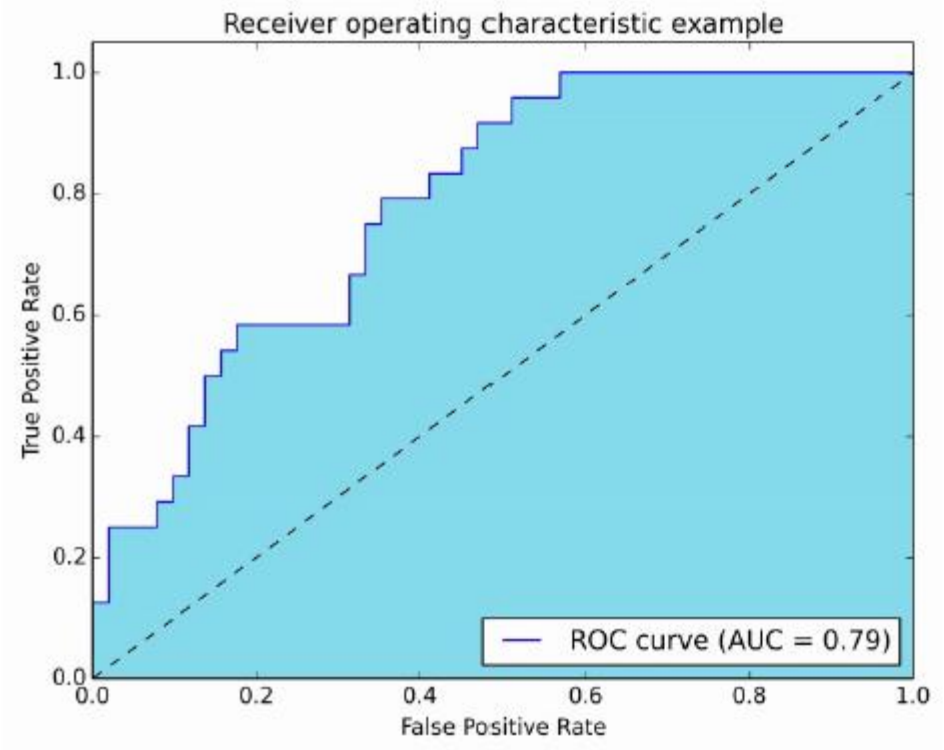
8	4
92	96

Threshold	TP	FP	TN	FN	TPR	FPR
0,99	1	0	100	99	0,01	0
0,98	2	0	100	98	0,02	0
0,96	2	1	99	98	0,02	0,01
...	...	...	...	...	...	...
0,65	8	4	96	92	0,04	0,08

ROC CURVE


Confusion matrix
ROC point from matrix
ROC graph
Instance
Class
Score
0.65
0.71
0.74
0.75
0.80
0.84
0.85
0.87
0.88
0.90
0.96
0.98
0.99
(0.04, 0.08)
(0.01, 0.02)
(0, 0.02)
(0, 0.01)
(0, 0)
0.10
0.08
0.06
0.04
0.02
0
0.02
0.04
0.06
0.08
0.10

ROC CURVE



http://www.navan.name/roc/

AUC (AREA UNDER THE ROC CURVE)

Practical Application in Medicine

- **Evaluating Diagnostic Tests**: A higher AUC indicates better performance in distinguishing between patients with and without a condition.
- **Model Comparison**
- **Threshold Selection**

DISCUSSION

Low Precision, High Recall

Goal: Catch **every possible case**, even if it means some false positives.
Why: Missing a true case (false negative) could lead to **disease progression**, **transmission**, or **death**.
Example: **Screening for highly contagious diseases** like **tuberculosis** or **HIV**.

- These tests are designed to be **highly sensitive** (high recall) to ensure no case is missed.
- False positives are acceptable because confirmatory testing can follow.

High Precision, Low Recall

Goal: Only flag cases when **very confident**, even if some true cases are missed.
Why: False positives may lead to **invasive**, **risky**, or **costly** follow-up procedures.
Example: **Prostate cancer screening** using the **PSA test**.

- PSA tests can lead to **overdiagnosis** and **unnecessary biopsies**.
- In older patients, where slow-growing cancers may not impact life expectancy, **precision is prioritized** to avoid overtreatment

CASE-STUDY

DECISION MAKING

original reports

Developing a Prediction Model for Pathologic Complete Response Following Neoadjuvant Chemotherapy in Breast Cancer: A Comparison of Model Building Approaches

Robert B. Basmadjian, MSc[1]; Shiying Kong, MSc[1,2,3]; Devon J. Boyne, PhD[1,2]; Tamer N. Jarada, PhD[2], Yuan Xu, MD, PhD[1,2,3]; Winson Y. Cheung, MD[1,2]; Sasha Lupichuk, MD[1,2]; May Lynn Quan, MD[1,2,3]; and Darren R. Brenner, PhD[1,2]

https://doi.org/10.1200/CCI.21.00055

EXERCISE

- notebook

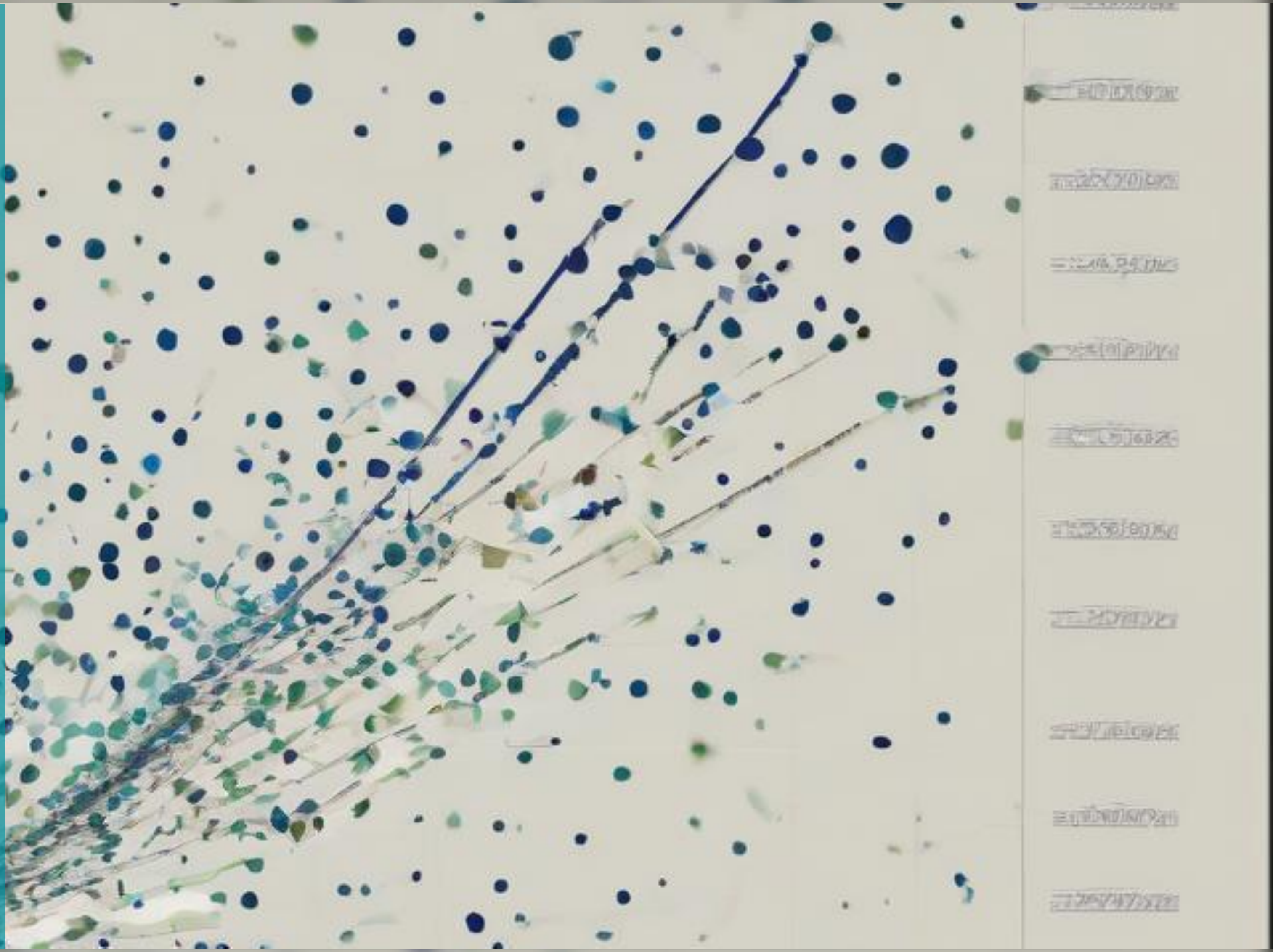
METRIC SELECTION

Some considerations when selecting the right metric for the classification task:

- Binary vs. Multiclass Classification
- Imbalanced Classes
- Objective of the Model (minimizing false positives or negatives)
- Threshold-Independent or -Dependent Metrics
- Interpretability
- Clinical Utility

02

MODEL EVALUATION METRICS FOR REGRESSION



INTRODUCTION TO MODEL EVALUATION METRICS

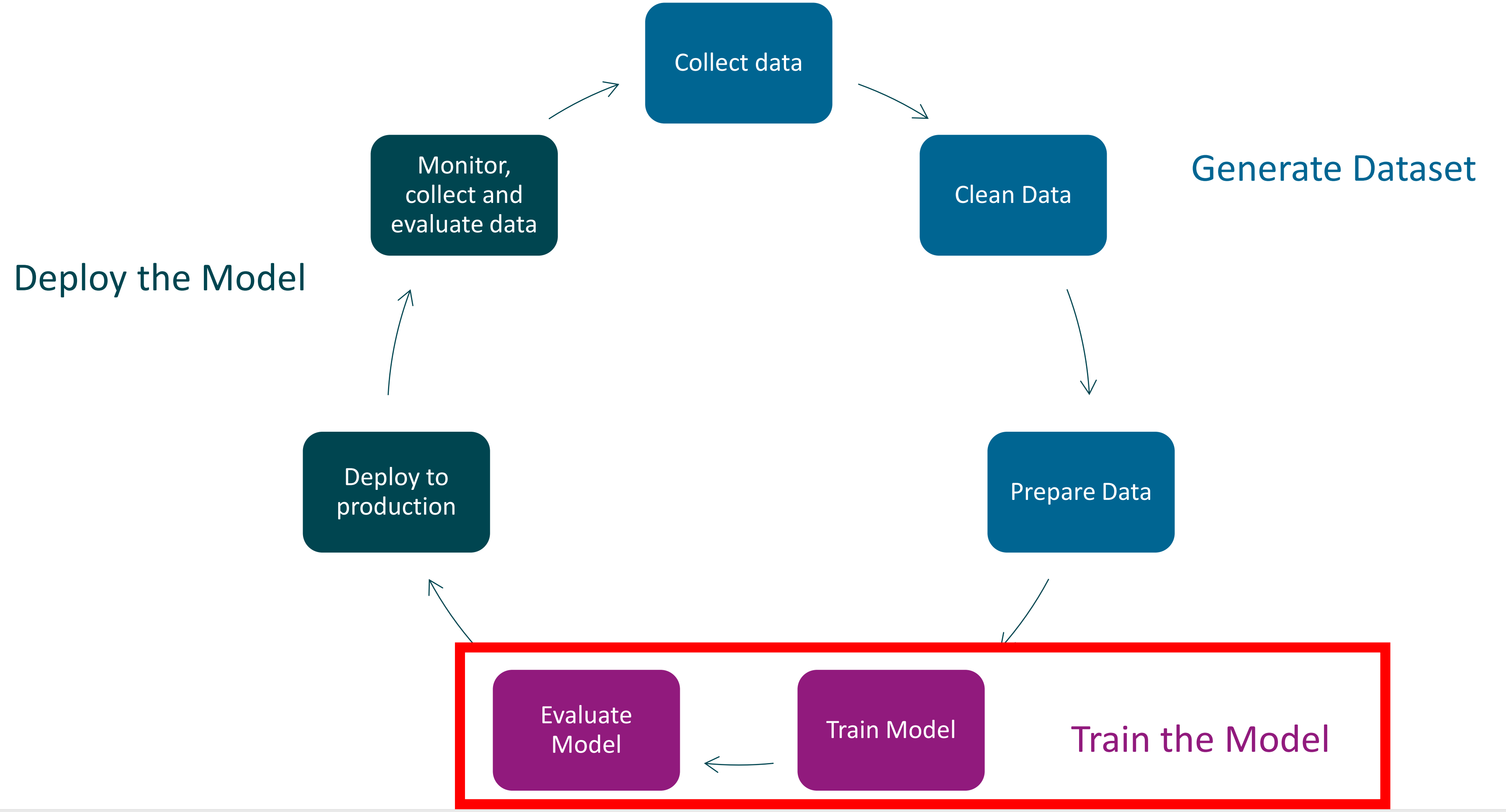
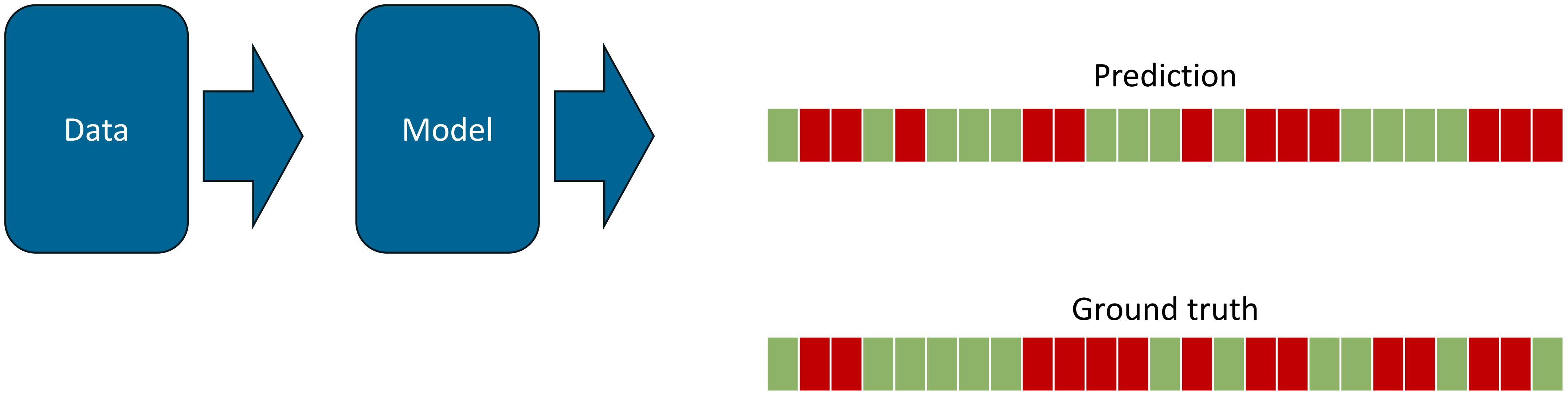


Image from: What is a Machine Learning Pipeline? - Datatron

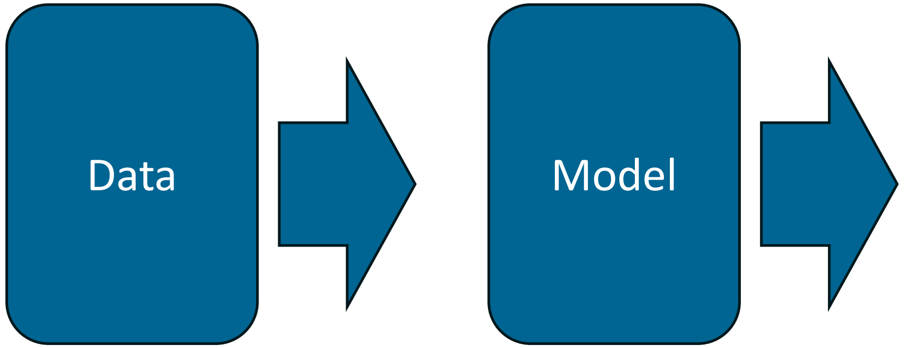
MODEL OUTPUT

Regression model



MODEL OUTPUT

Regression model



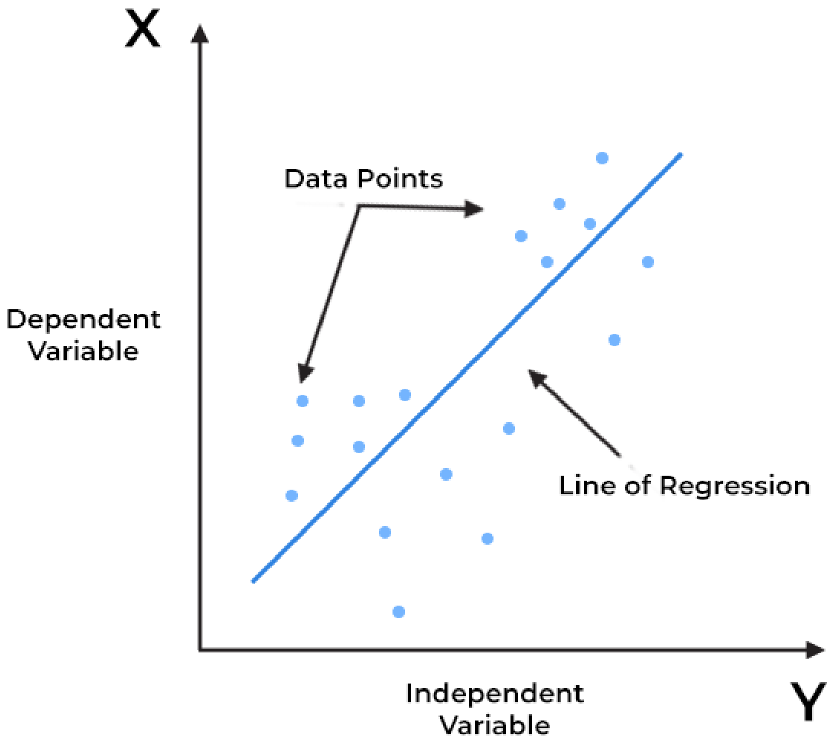


Image from: https://www.spiceworks.com/tech/artificial-intelligence/articles/what-is-linear-regression/

MODEL OUTPUT

Metrics

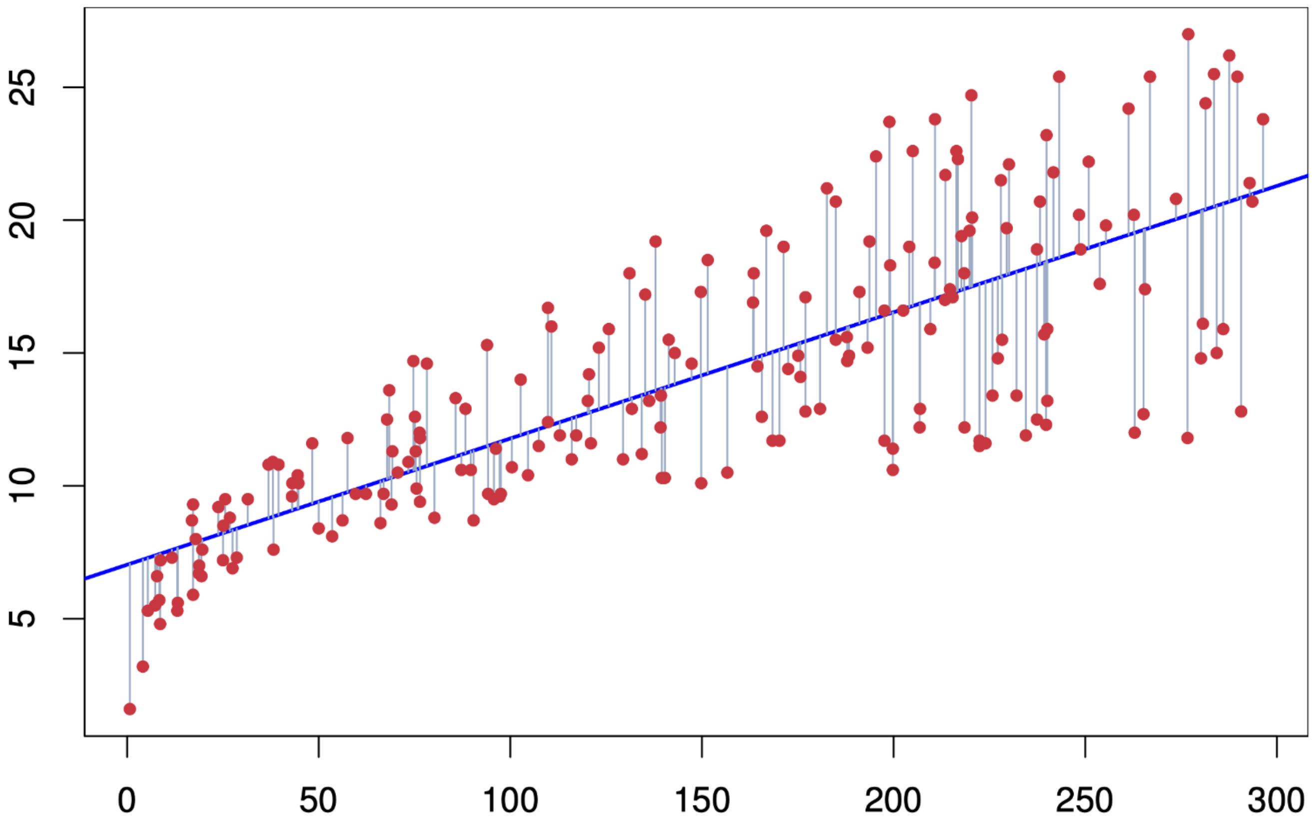


Image from: https://web.stanford.edu/class/stats202/notes/Linear-regression/Simple-linear-regression.html

MEAN ABSOLUTE ERROR (MAE)

measures the average magnitude of errors between predicted and actual values

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

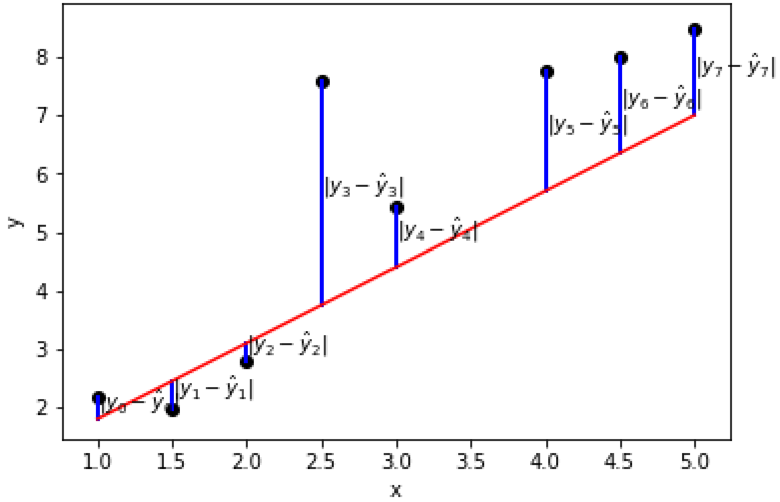


Image from: https://jmlb.github.io/flashcards/2018/07/01/mae_vs_rmse/

MEAN SQUARED ERROR (MSE)

Importance

- **Interpretability**: MAE is easy to understand and interpret, as it represents the average error in the same units as the target variable.
- **Robustness to Outliers**: Unlike metrics such as Mean Squared Error (MSE), MAE is less sensitive to outliers because it does not square the error terms.

Limitations

- **Equal Weight to All Errors**: MAE gives equal weight to all errors, regardless of their magnitude. This can be a limitation in cases where larger errors need to be penalized more.

EXERCISE

- Y = 1, 3, 5
- Y_pred = 0.5, 3,5, 5

ROOT MEAN SQUARED ERROR (RMSE)

measures the square root of the average of the squared differences between predicted and actual values

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

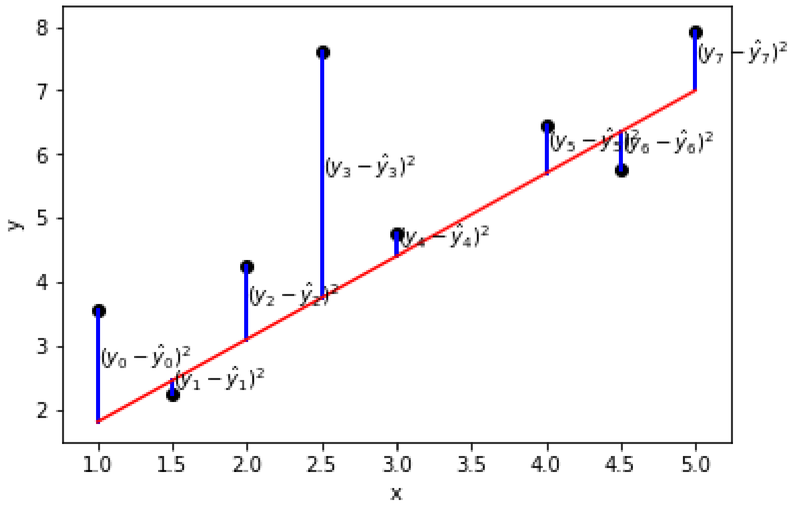


Image from: https://jmlb.github.io/flashcards/2018/07/01/mae_vs_rmse/

ROOT MEAN SQUARED ERROR (RMSE)

Importance

- **Sensitive to Large Errors**: RMSE is more **sensitive to large errors** compared to MAE because the errors are squared before averaging. This means that RMSE can be particularly useful when large errors are undesirable.

Limitations

- **Sensitivity to Outliers**: Because RMSE squares the error terms, it can be **disproportionately influenced by outliers**. Large errors will have a greater impact on RMSE.
- **Comparability**: RMSE values are context-specific and can only be used to compare models on the same dataset or problem.

R-SQUARED (R2)

statistical measure that represents the **proportion of the variance** for a dependent variable that's explained by an independent variable or variables in a regression model

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

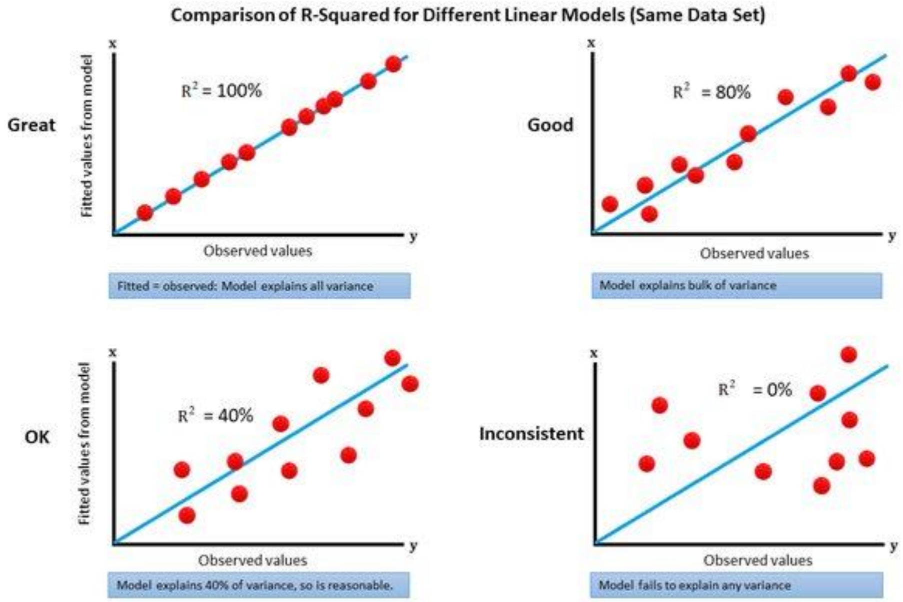
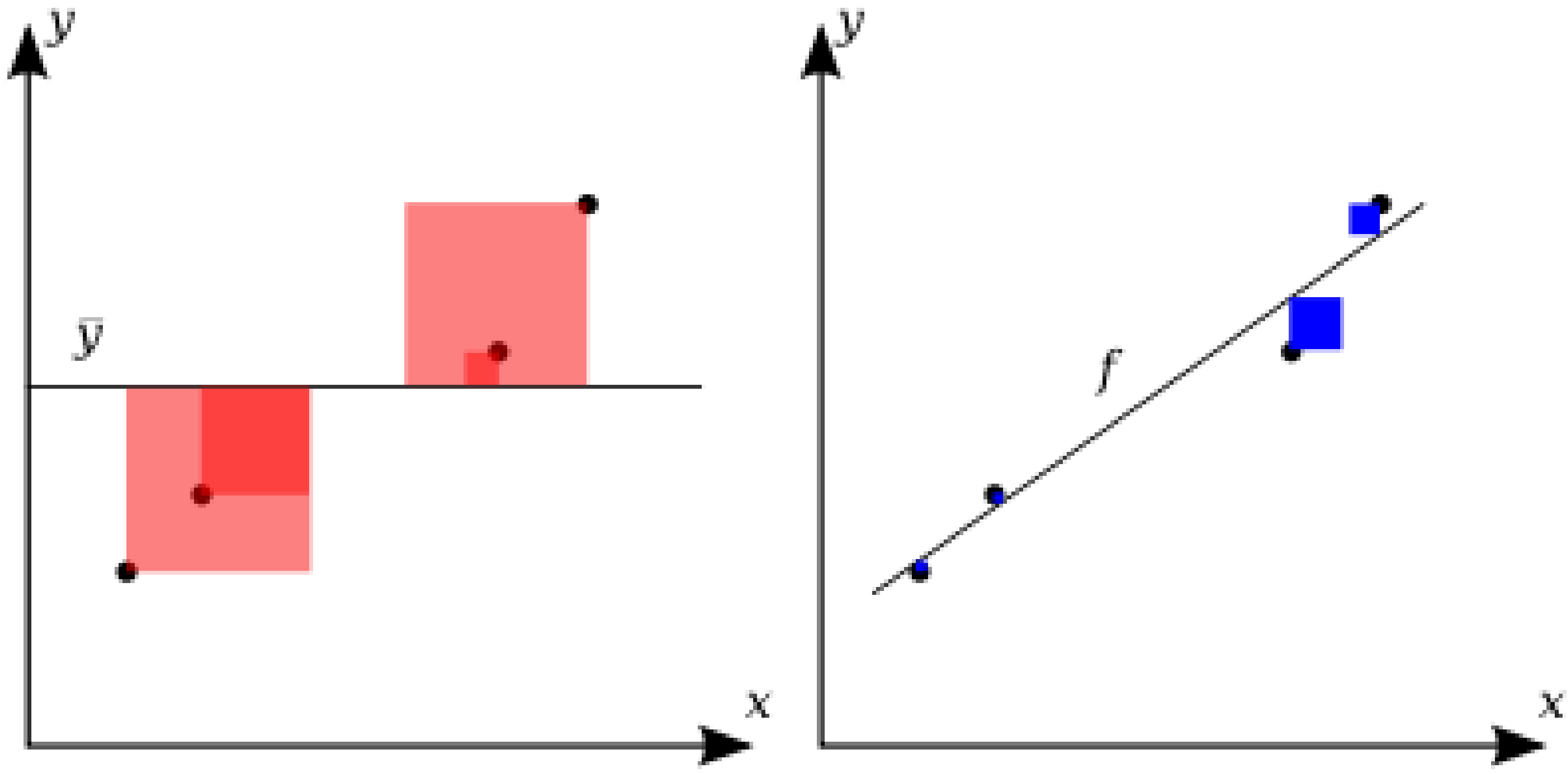


Image from: https://www.datasciencecentral.com/r-squared-in-one-picture/

R-SQUARED (R2)



$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

Image from:
https://en.wikipedia.org/wiki/Coefficient_of_determination#/media/File:Coefficient_of_Determination.svg

R-SQUARED (R2)

Importance

- **Goodness of Fit**: R^2 is a key indicator of the goodness of fit of a model. It shows how well the model explains the variability in the dependent variable.
- **Model Comparison**: R^2 can be used to compare the explanatory power of different models for the same dataset.

Limitations

- **Does Not Indicate Causation**: A high R^2 value does not imply that the model variables are causally related to the dependent variable.
- **Not Suitable for Non-Linear Models**: R^2 is primarily used for linear regression models and may not be appropriate for evaluating non-linear models.

MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)

expresses the average absolute error
between the predicted and actual values
as a percentage of the actual values

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)

Importance

- **Interpretability**: MAPE is **easy to understand and interpret**, as it represents the error as a percentage of the actual values.
- **Normalization**: MAPE provides a normalized measure of error, making it useful for comparing model performance across different datasets and scales.

Limitations

- **Division by Zero**: MAPE is undefined if any actual value (yi) is zero, as this would involve division by zero.
- **Sensitivity to Small Values**: MAPE can be excessively high if the actual values are very small, as small differences between predicted and actual values result in large percentage errors.

03

TRAIN-TEST SPLIT AND CROSS VALIDATION

INTRODUCTION TO TRAIN-TEST SPLIT

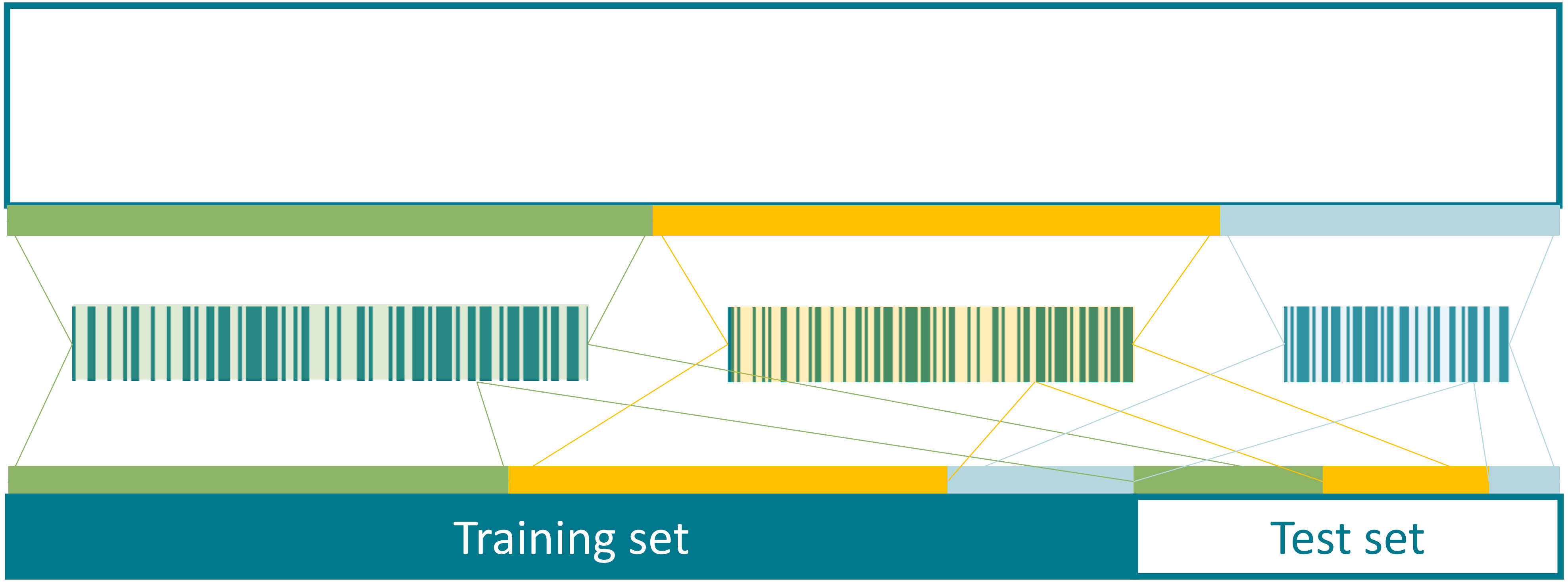
Models should be robust, generalizable, and capable of performing well on unseen data

TRAIN-TEST SPLIT

Data set

RANDOM SPLIT

STRATIFIED SPLIT



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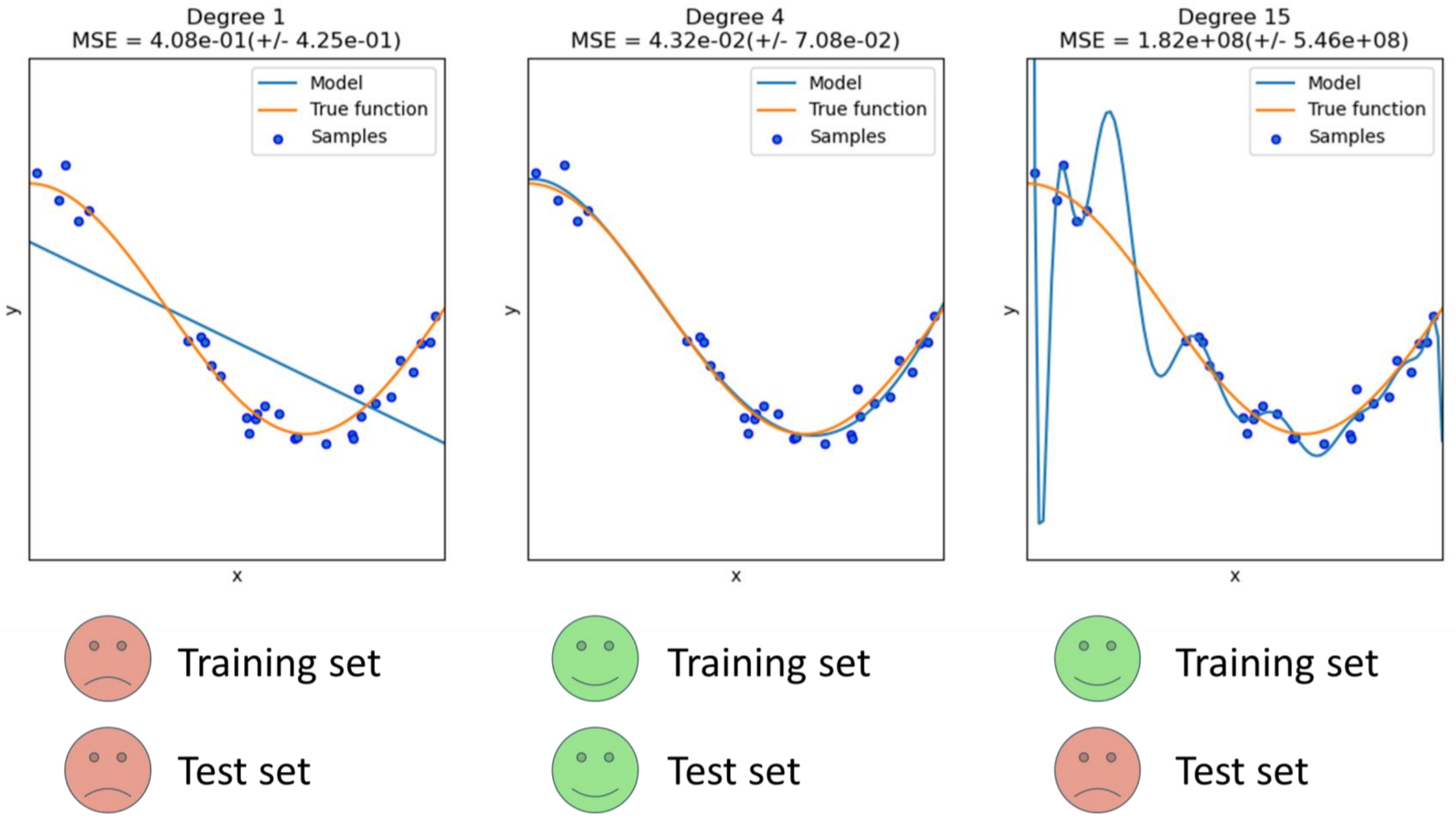
OVERFITTING AND UNDERFITTING



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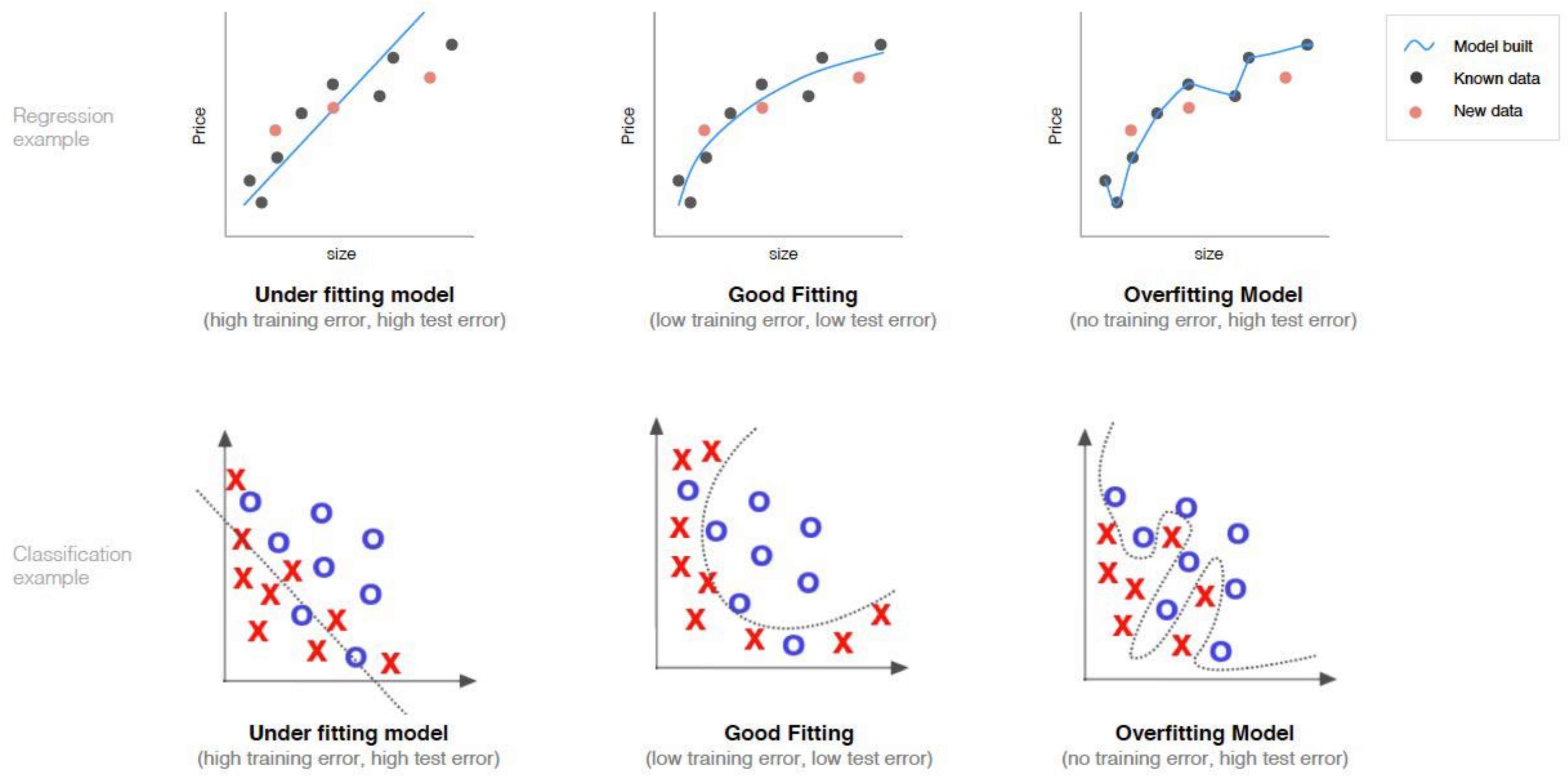
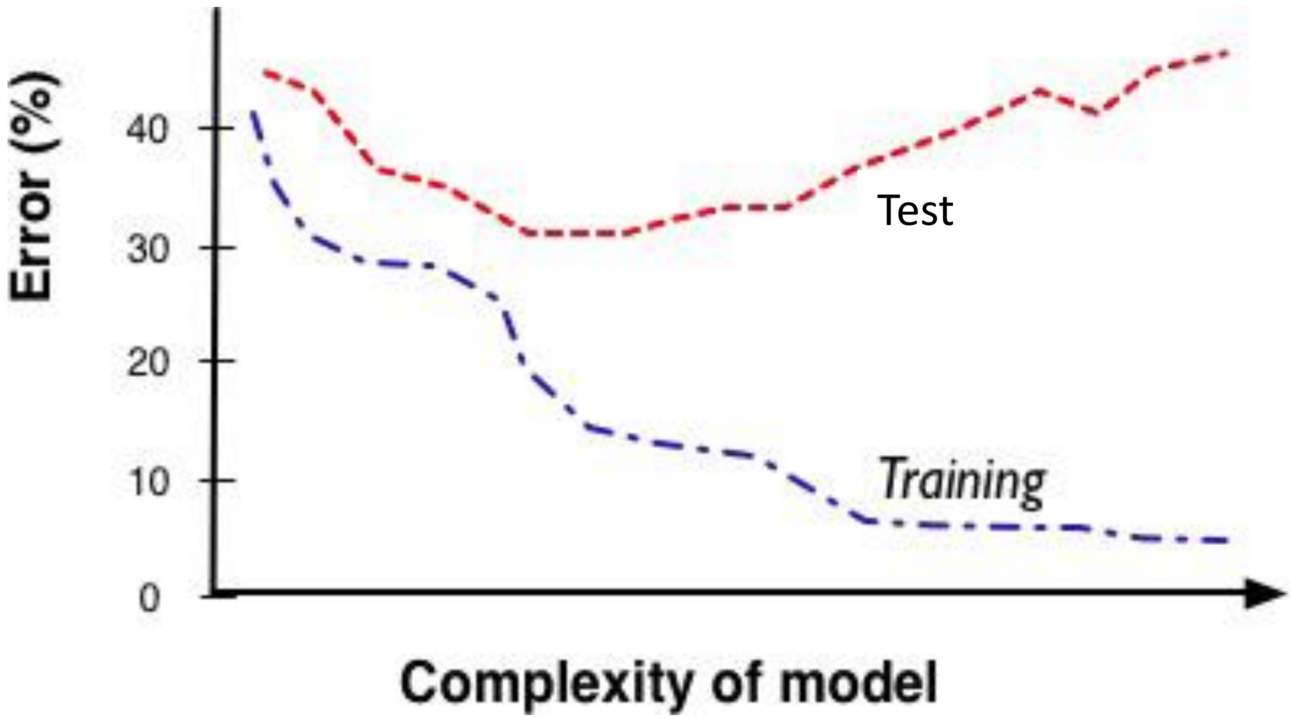
OVERFITTING AND UNDERFITTING



OVERFITTING AND UNDERFITTING



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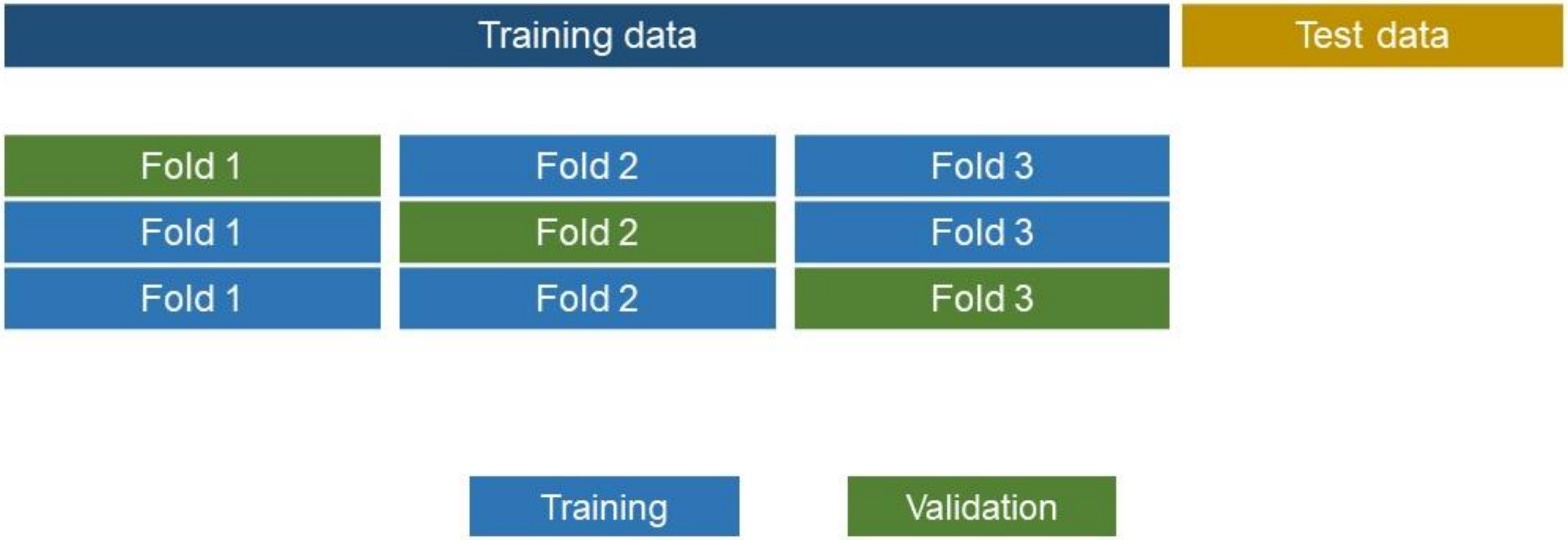
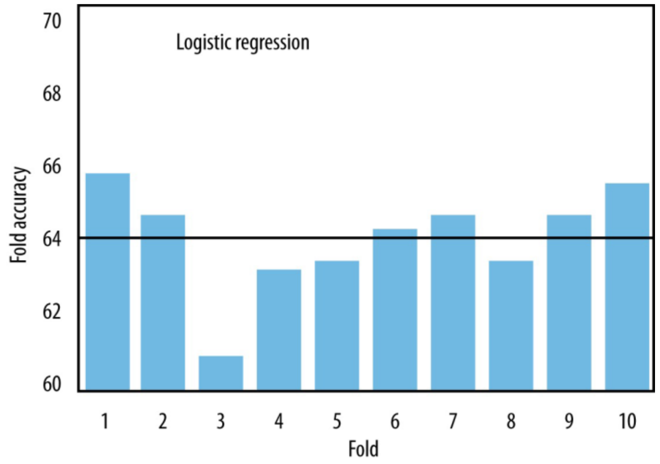
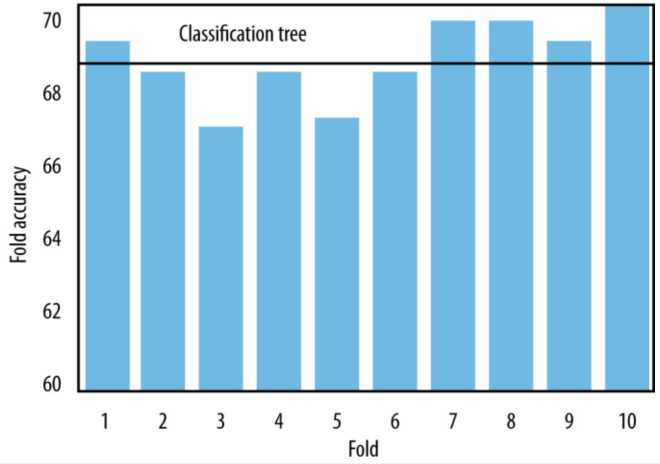
K-FOLD CROSS-VALIDATION



K-FOLD CROSS-VALIDATION





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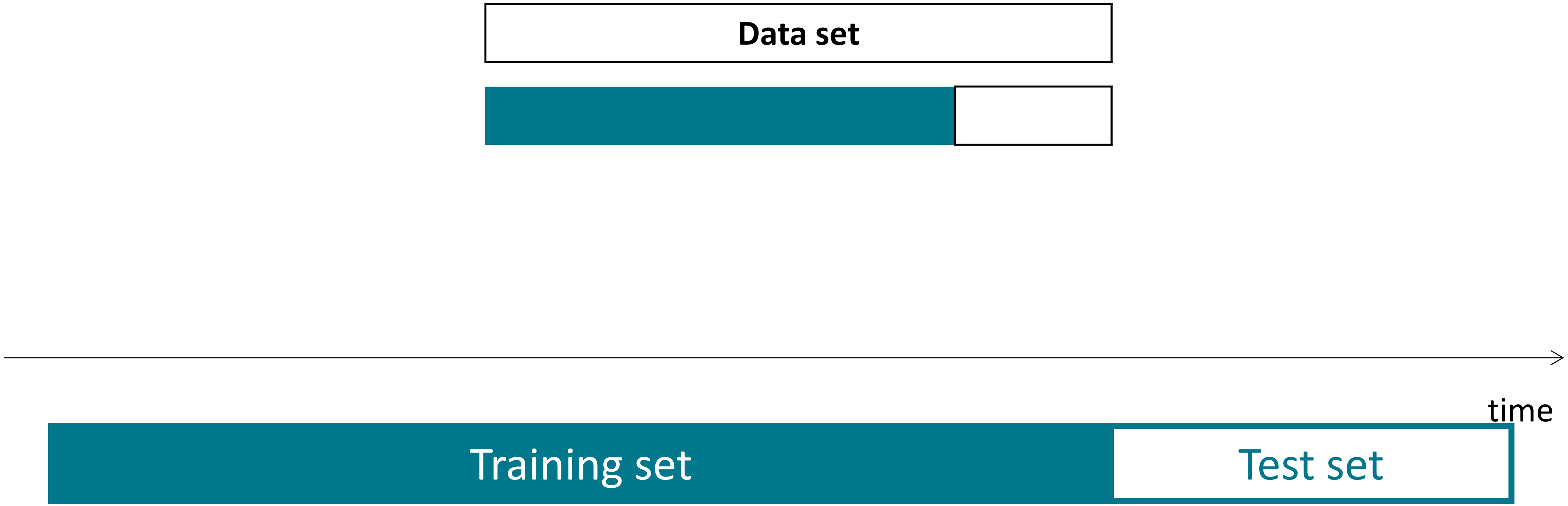
TIME-SERIES SPLIT

(for medical data with temporal component)



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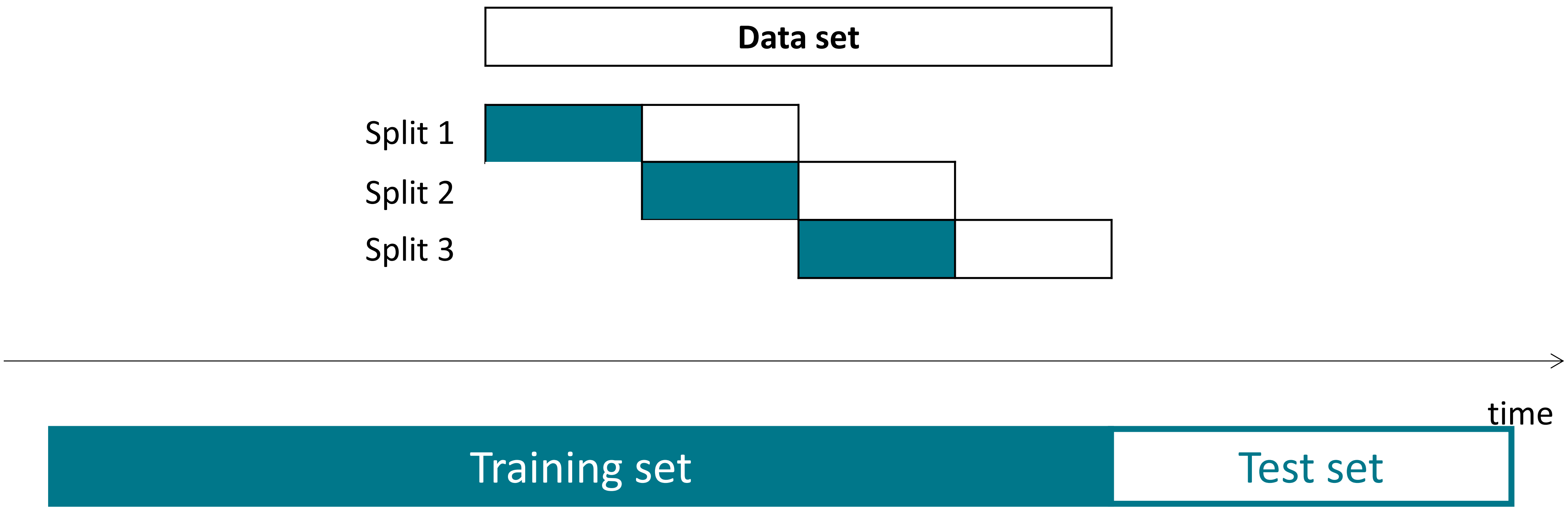
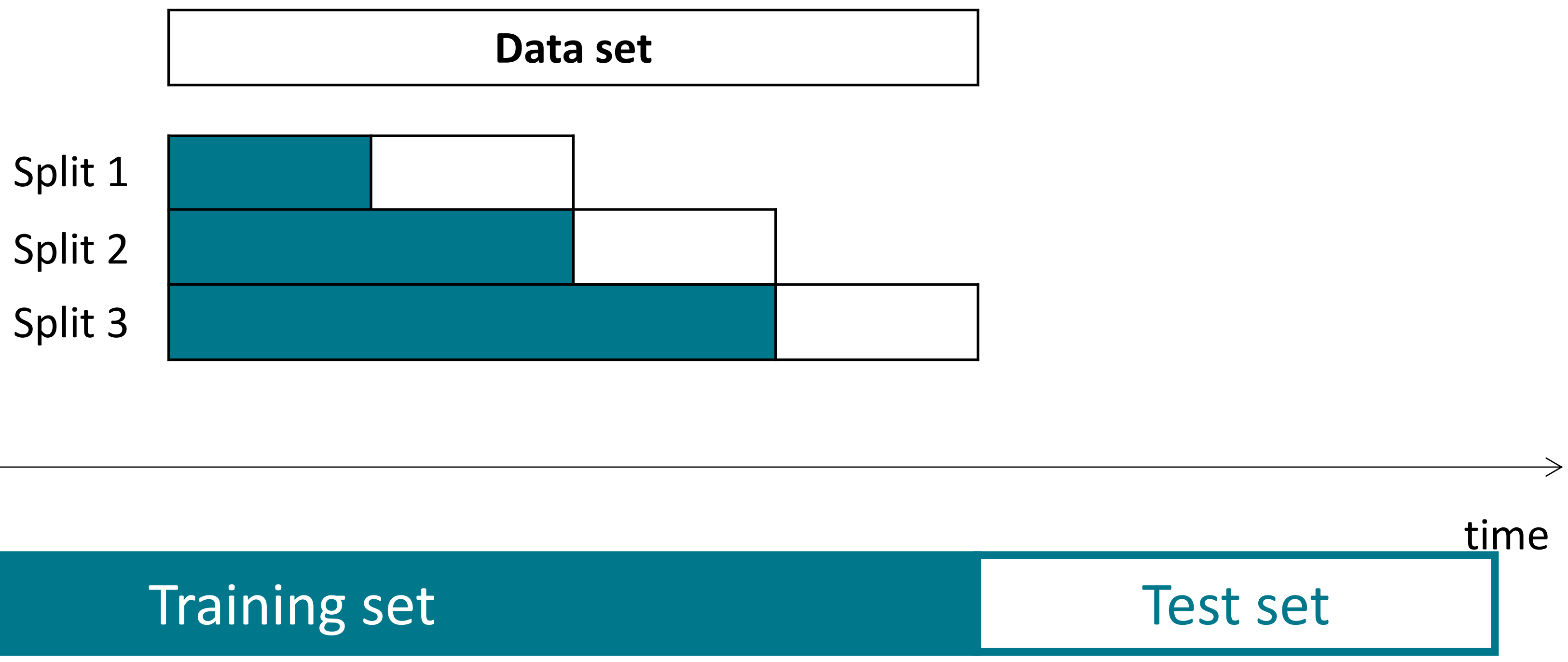
TIME-SERIES SPLIT – SLIDING WINDOW

(for medical data with temporal component)



TIME-SERIES SPLIT – EXPANDING WINDOW

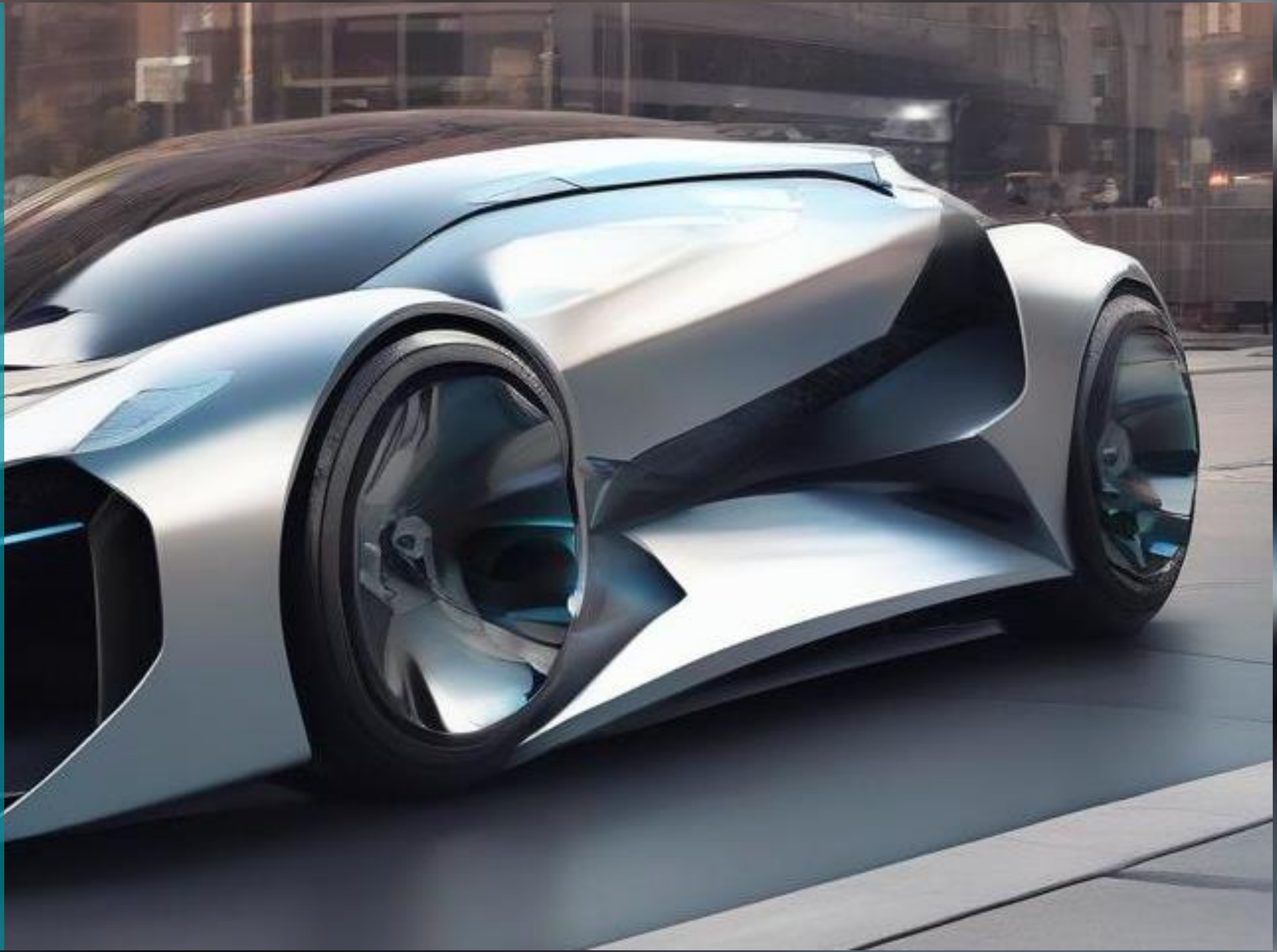
(for medical data with temporal component)



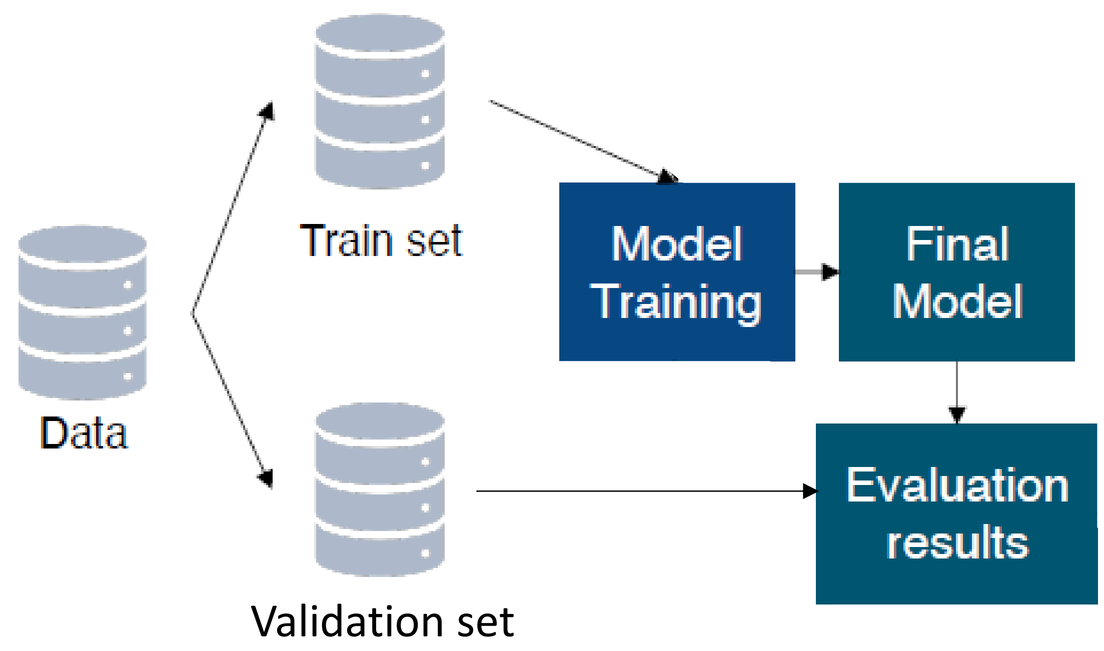
EXERCISE

04

MODEL TUNING AND OPTIMIZATION



INTRODUCTION TO MODEL TUNING



HYPERPARAMETERS AND PARAMETERS

Parameters	Hyperparameters
• Estimated during the training • Its part of the model • The estimated value is saved with the trained model to make prediction Model Training	• Values are set beforehand • External to the model • Not a part of the trained model and hence the vales are not saved for making prediction Hyperparameter tuning

HYPERPARAMETERS AND PARAMETERS

LOGISTIC REGRESSION

$$\hat{y} = \sigma(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n)$$

```
class sklearn.linear_model.LogisticRegression(penalty='l2', *, dual=False,
tol=0.0001, C=1.0, fit_intercept=True, intercept_scaling=1, class_weight=None,
random_state=None, solver='lbfgs', max_iter=100, multi_class='deprecated', verbose=0,
warm_start=False, n_jobs=None, l1_ratio=None)
```

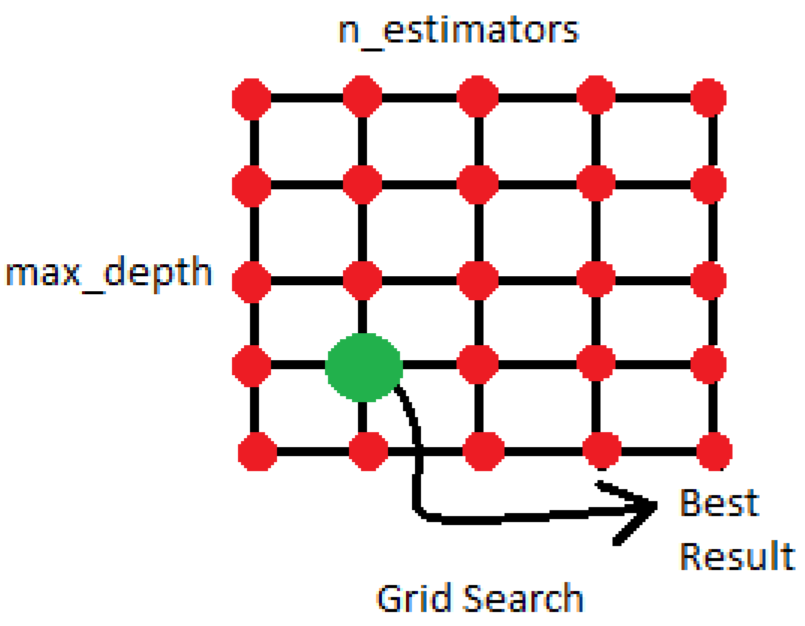
[source]

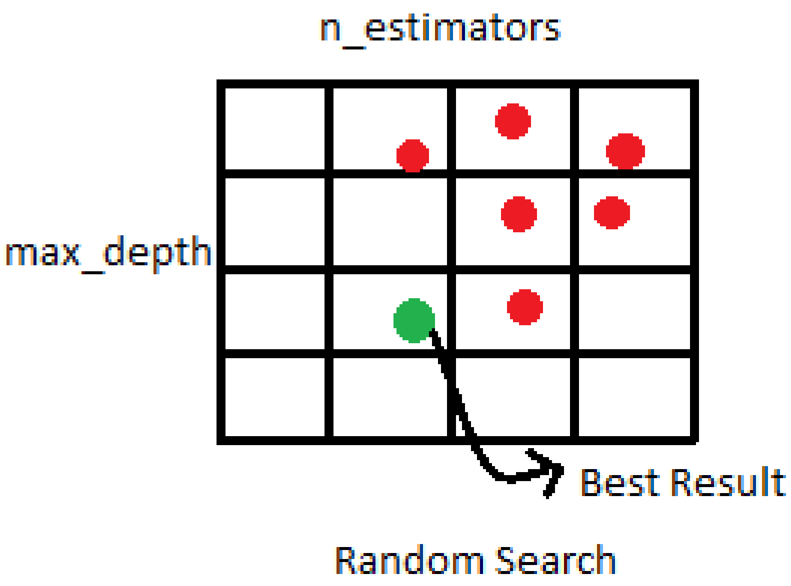
HYPERPARAMETERS AND PARAMETERS

DECISION TREE

```
# Create a DecisionTreeClassifier object
dt_classifier = DecisionTreeClassifier(
    criterion='gini',          # Split criterion
    max_depth=5,               # Maximum depth of the tree
    min_samples_split=10,      # Minimum number of samples required to split an internal node
    min_samples_leaf=5,        # Minimum number of samples required to be at a leaf node
    max_features='sqrt',       # Number of features to consider when looking for the best spl
    max_leaf_nodes=20,         # Maximum number of leaf nodes
    min_impurity_decrease=0.01,  # Minimum impurity decrease required to split a node
    random_state=42            # Random state for reproducibility
)
```

GRID SEARCH VS. RANDOM SEARCH



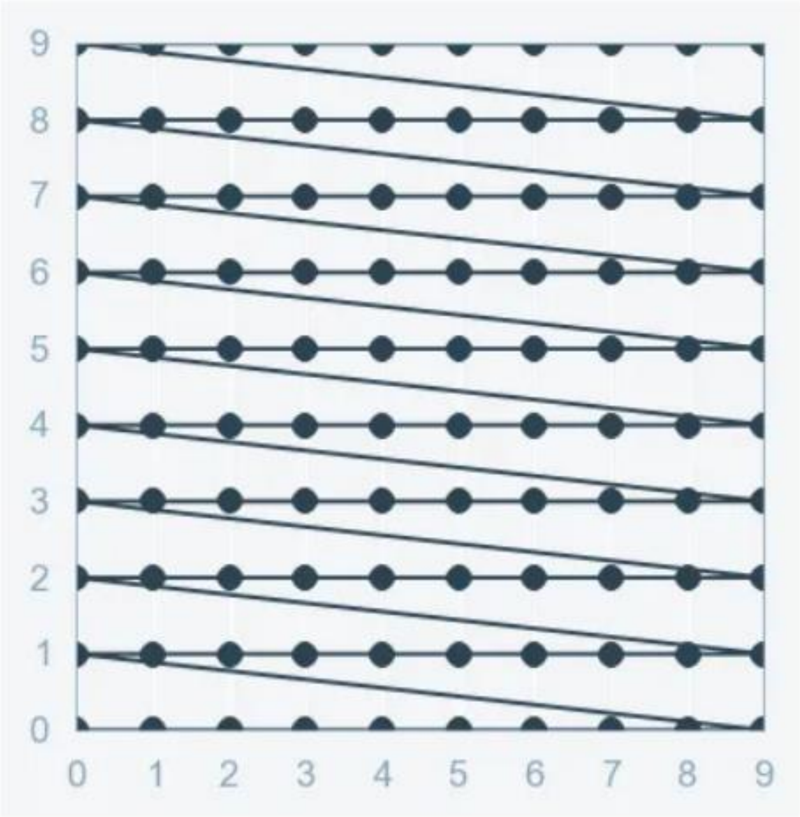


GRID SEARCH VS. RANDOM SEARCH

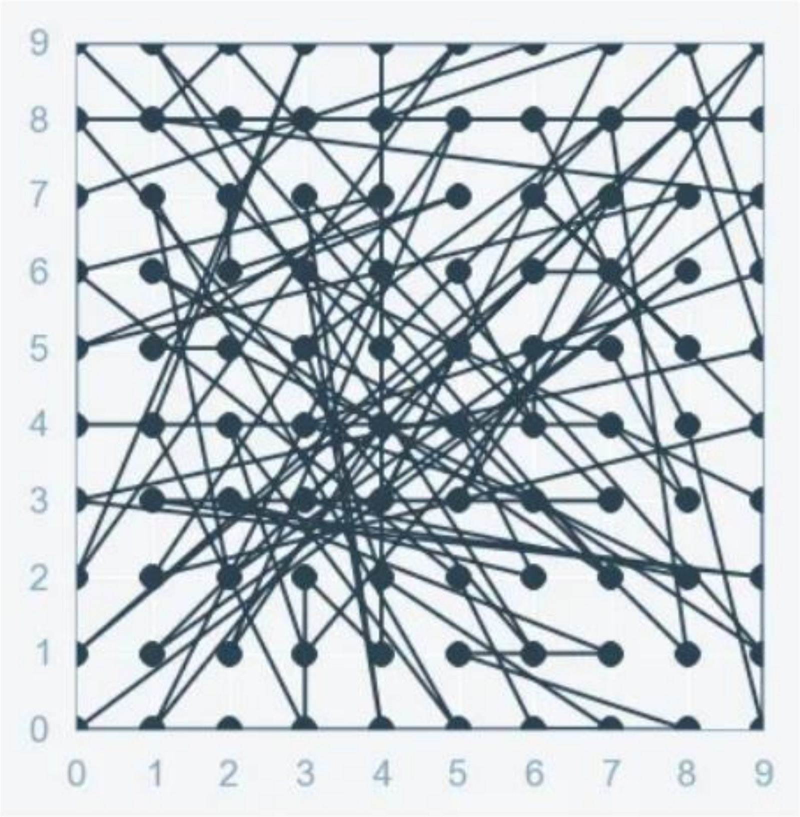
C \ Alpha	0.1	0.2	0.3	0.4
0.5	0.701	0.703	0.697	0.696
0.4	0.699	0.702	0.698	0.702
0.3	0.721	0.726	0.713	0.703
0.2	0.706	0.705	0.704	0.701
0.1	0.698	0.692	0.688	0.675

GRID SEARCH VS. RANDOM SEARCH

GRID SEARCH



RANDOM SEARCH



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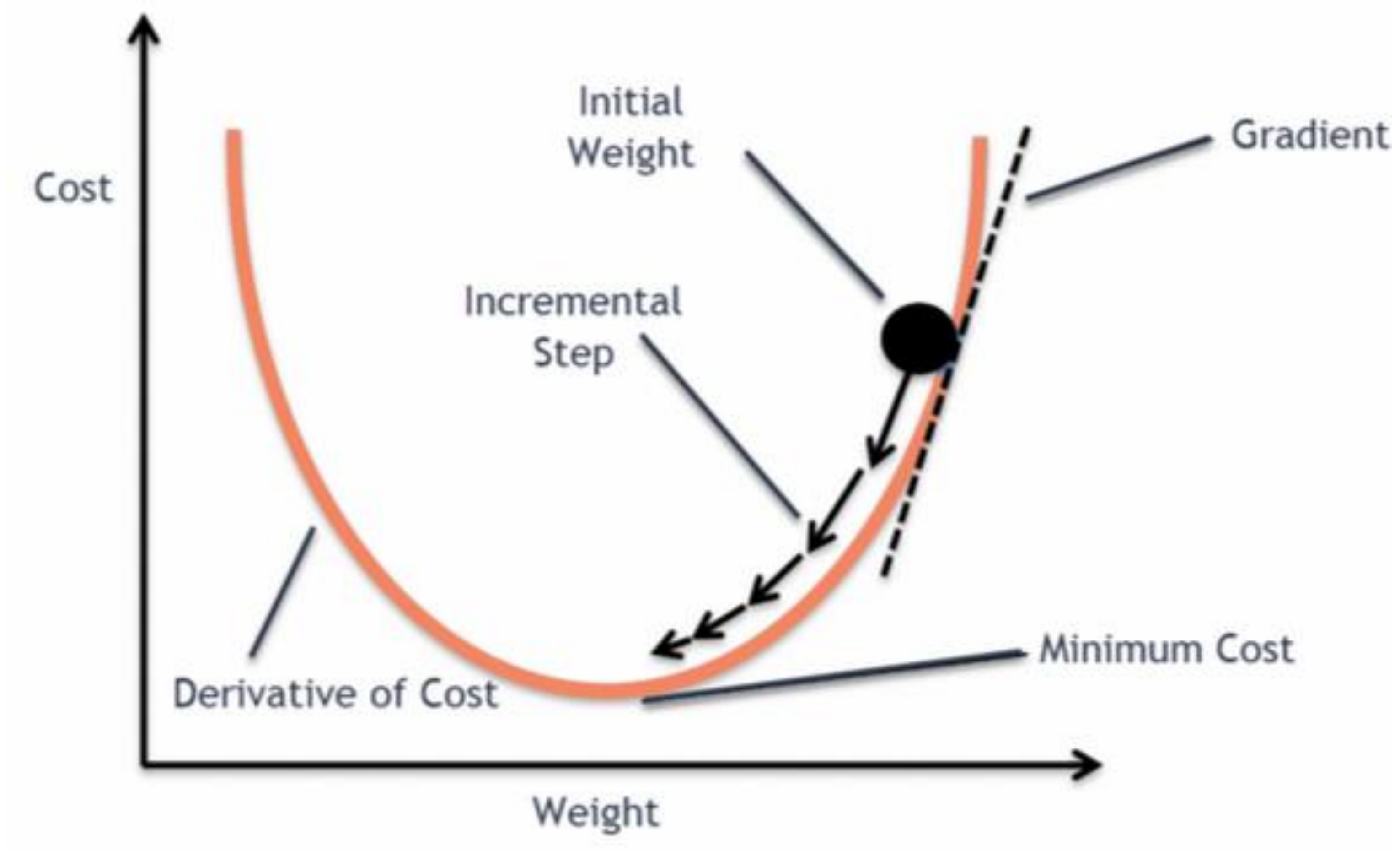
EXERCISE

- GRID SEARCH

GRADIENT DESCENT



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EXERCISE

- GRADIENT DESCENT

GRADIENT DESCENT UPDATE RULE

LOGISTIC REGRESSION

$$\beta_j = \beta_j - \alpha \frac{\partial \text{Log Loss}}{\partial \beta_j}$$

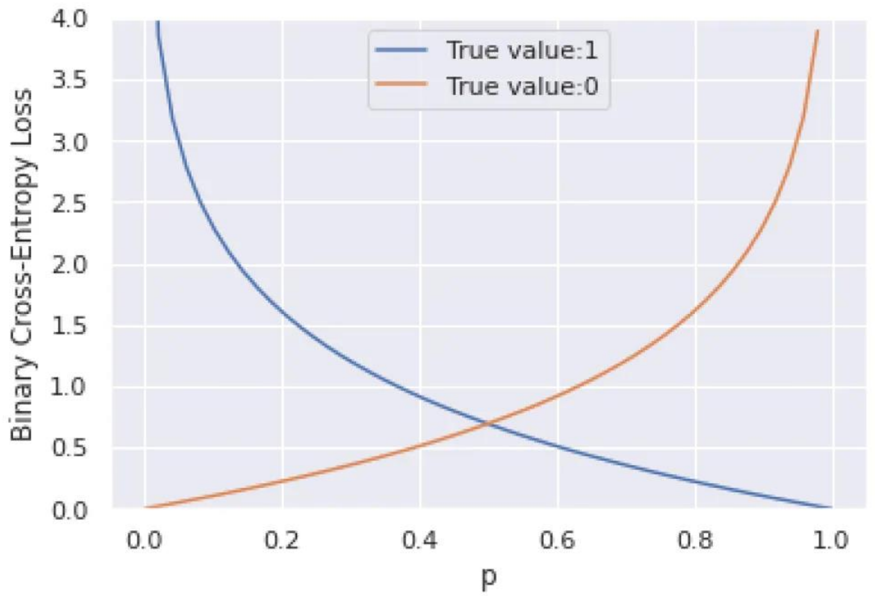
$$\hat{y} = \sigma(\beta_0 + \beta_1 X_1 + \beta_2 X_2)$$

LOG LOSS

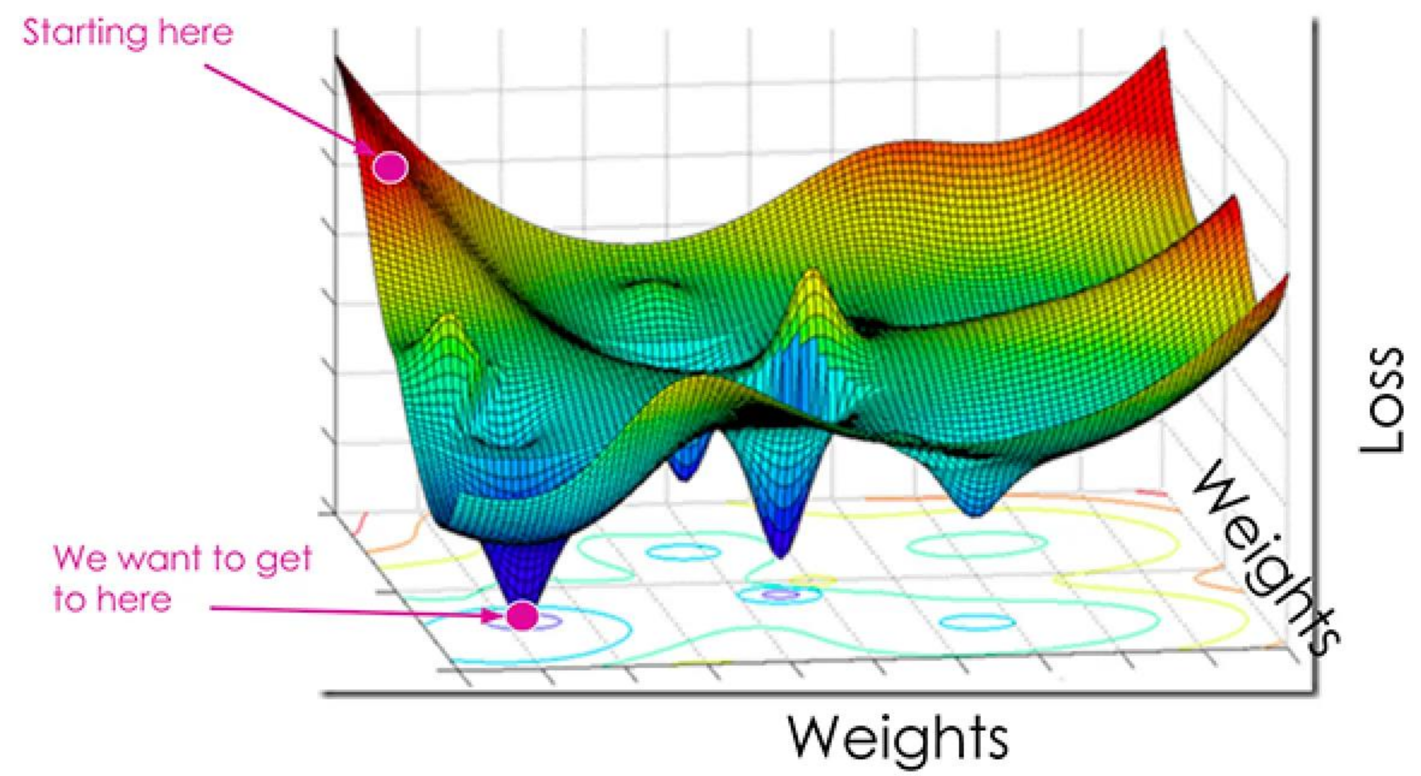
$$\text{Log Loss} = -\frac{1}{m}\sum_{i=1}^{m}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right]$$

$$\beta_j = \beta_j - \alpha \frac{\partial \text{Log Loss}}{\partial \beta_j}$$

$$\hat{y} = \sigma(\beta_0 + \beta_1 X_1 + \beta_2 X_2)$$



OPTIMIZATION



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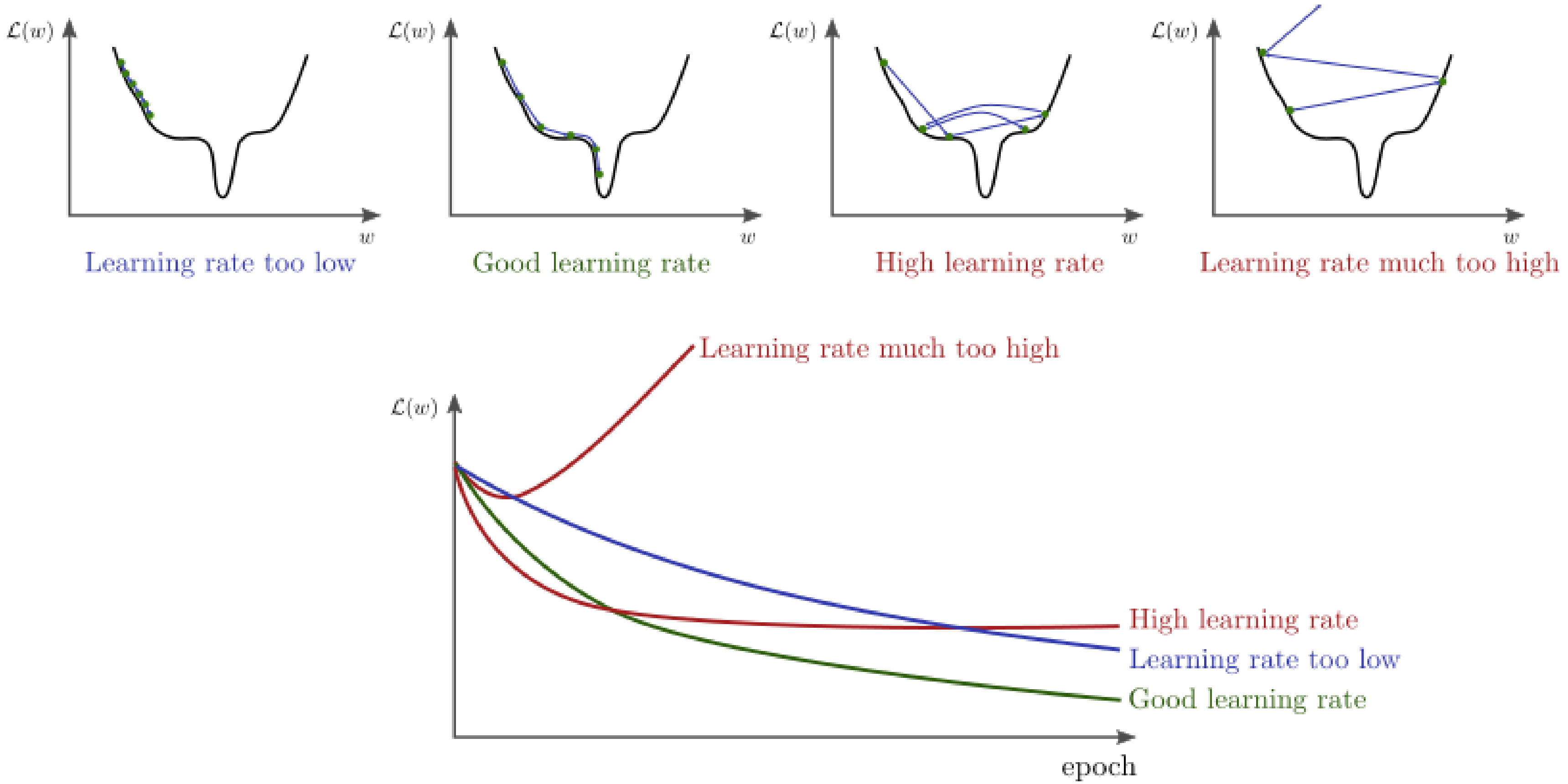
LEARNING RATE

$$\beta_j = \beta_j - \alpha \frac{\partial \text{Log Loss}}{\partial \beta_j}$$



From https://www.bdhammel.com/learning-rates/

REGULARIZATION TECHNIQUES

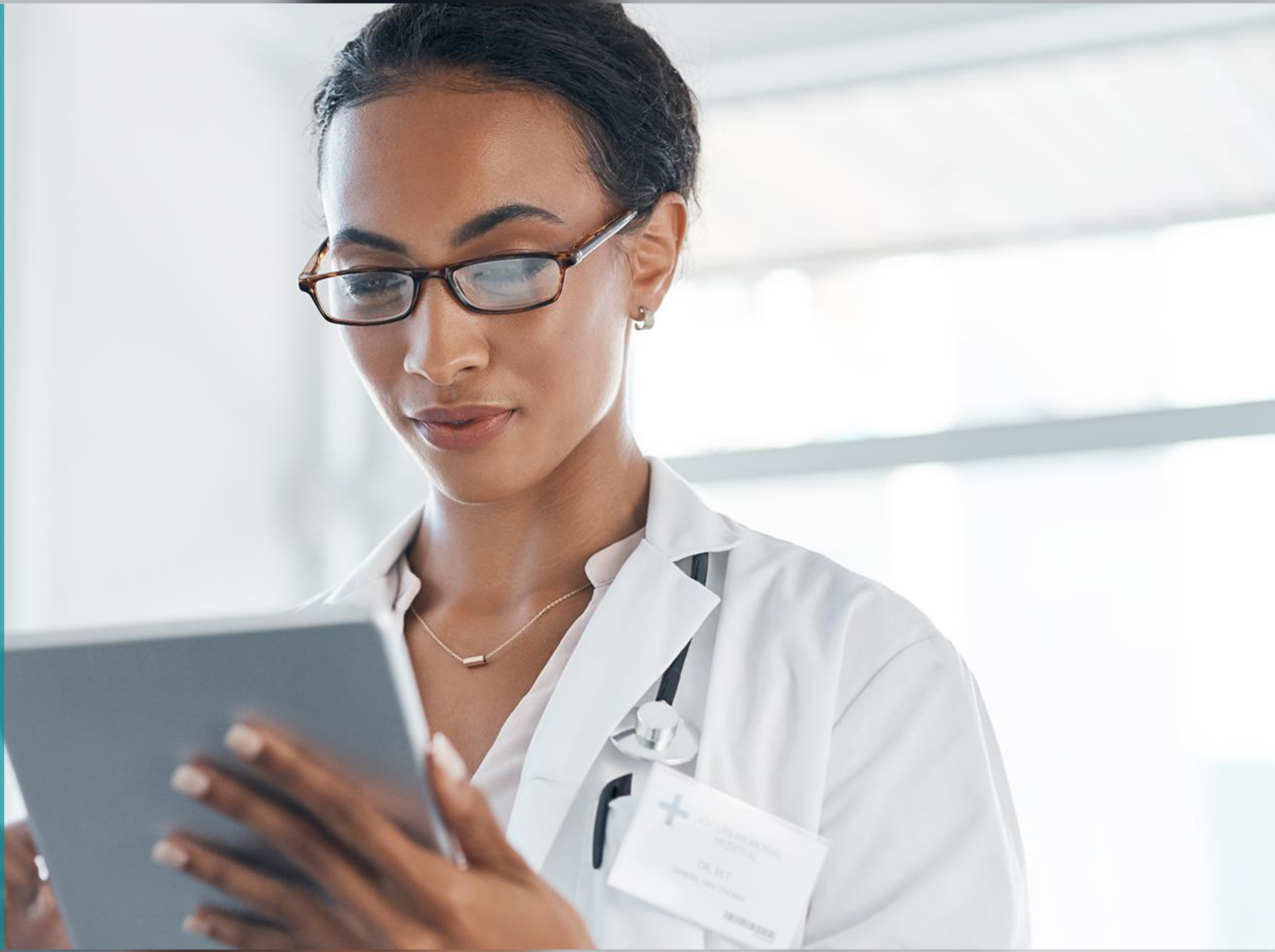
$$\text{Log Loss} = -\frac{1}{m}\sum_{i=1}^{m}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right]$$

$$\text{Log Loss} + \text{L2 Regularization} = -\frac{1}{m}\sum_{i=1}^{m}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] + \frac{\lambda}{2m}\sum_{j=1}^{n}\beta_j^2$$

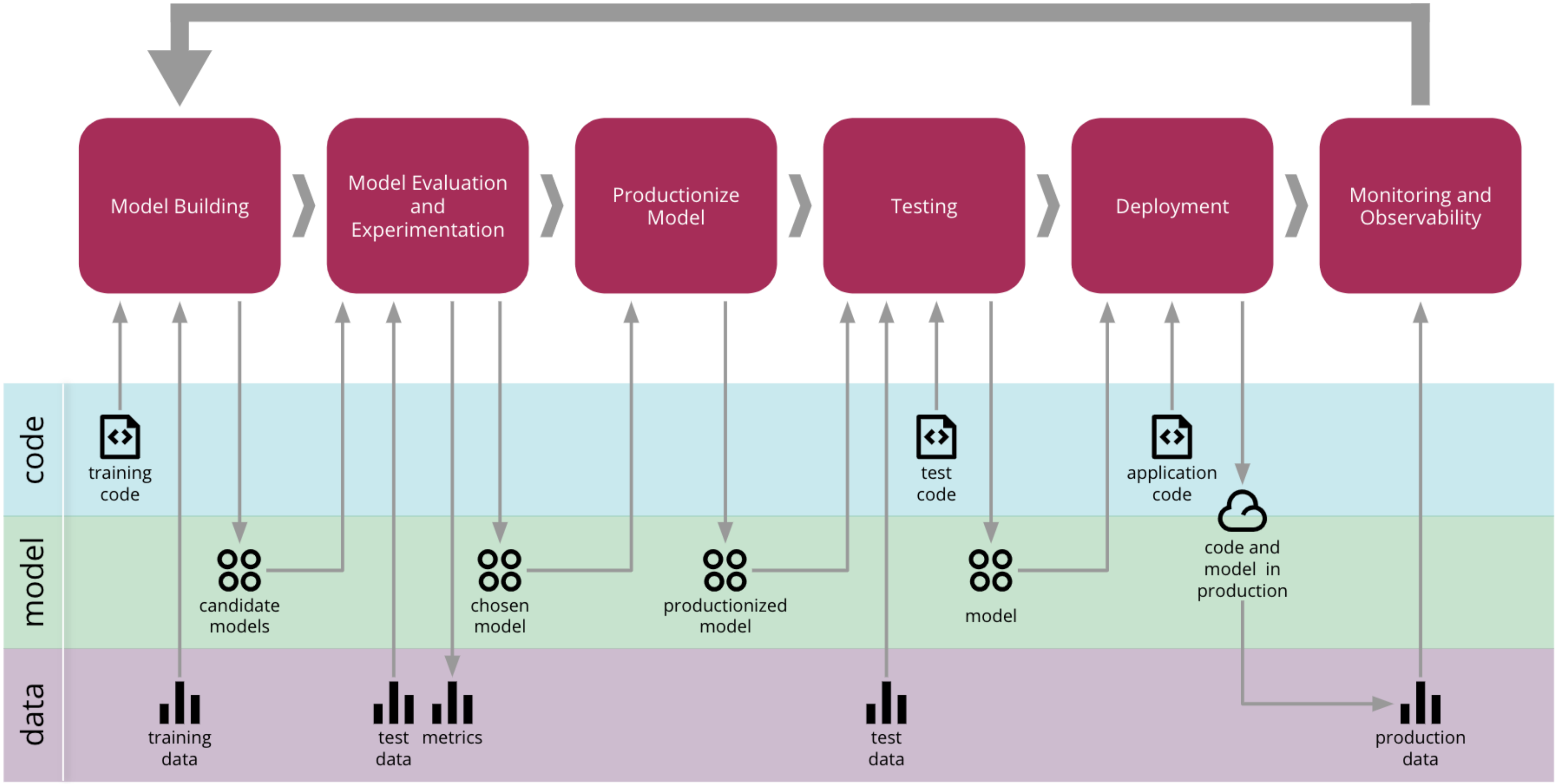
$$\text{Log Loss} + \text{L1 Regularization} = -\frac{1}{m}\sum_{i=1}^{m}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] + \frac{\lambda}{m}\sum_{j=1}^{n}|\beta_j|$$

$$\text{Log Loss} + \text{Elastic Net Regularization} = -\frac{1}{m}\sum_{i=1}^{m}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] + \frac{\lambda_1}{m}\sum_{j=1}^{n}|\beta_j| + \frac{\lambda_2}{2m}\sum_{j=1}^{n}\beta_j^2$$

CONCLUSION



MODEL MONITORING



From https://martinfowler.com/articles/cd4ml.html

THANK YOU

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